

Ketamine: Drug information - UpToDate

Ketamine: Drug information - UpToDate

Official reprint from UpToDate®

www.uptodate.com © 2026 UpToDate, Inc. and/or its affiliates. All Rights Reserved.

Ketamine: Drug information

2026© UpToDate, Inc. and its affiliates and/or licensors. All Rights Reserved.

Contributor Disclosures

For additional information see "Ketamine: Patient drug information" and "Ketamine: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Ketalar

Brand Names: Canada

- Ketalar

Pharmacologic Category

- Antidepressant;
- General Anesthetic

Dosing: Adult

Dosage guidance:

Safety: To decrease risk of respiratory depression and apnea, administer IV bolus doses over >30 to 60 seconds (Ref).

Agitation, acute/severe or refractory

Agitation, acute/severe or refractory (adjunct) (off-label use): Note: May be used when benzodiazepines and/or antipsychotics have failed (Ref).

IV: Initial: 1 to 2 mg/kg once over 30 to 60 seconds; if initial sedation is inadequate, may repeat dose once 5 to 10 minutes after the initial dose using 0.5 to 1 mg/kg; however, this is not common if using the upper end of the initial dose range (Ref). Some experts use an initial dose of 0.5 to 1 mg/kg in the emergency department with concomitant sedatives (Ref).

IM: Initial: 4 to 6 mg/kg once; if initial sedation is inadequate, may repeat dose once 10 to 25 minutes after the initial dose using 2 to 3 mg/kg; however, this is not common if using the upper end of the initial dose range (Ref). Some experts use an initial dose of 2 mg/kg in the emergency department with concomitant sedatives (Ref). Maximum recommended dose: 400 mg (Ref).

Analgesia, subanesthetic dosing

Analgesia, subanesthetic dosing (off-label use):

Acute pain:

Note: Prior to use, consider consultation with a pain specialist or service experienced with ketamine use in this setting. Optimal doses and regimens have not been identified; refer to institutional protocols. Recommendations provided below are examples of regimens. May be useful for moderately to severely painful procedures and conditions that do not respond optimally to standard analgesics (eg, postoperative, burn, trauma, sickle cell disease vaso-occlusive pain).

IV intermittent infusion: 0.1 to 0.5 mg/kg once; usual dosage range: 10 to 30 mg. May be administered over 15 minutes or via slow IV push over 1 minute (Ref).

IV continuous infusion: Initial: 0.25 to 0.5 mg/kg bolus (maximum bolus: 35 mg), followed by 0.05 to 0.25 mg/kg/hour (or 0.8 to 4.2 mcg/kg/minute) continuous infusion; titrate to pain goal and tolerability; usual dosing range: 0.05 to 1 mg/kg/hour (or 0.8 to 16.7 mcg/kg/minute); may need to use doses at the higher range in patients who are opioid-tolerant or with opioid-induced hyperalgesia; duration of infusion: 48 to 72 hours (Ref).

Intranasal (off-label route): 0.2 to 1 mg/kg by administering half dose in each nostril (using 100 mg/mL solution); if necessary, may repeat after 10 to 15 minutes with 0.25 to 0.5 mg/kg; titrate to pain goal and tolerability. Doses up to 40 mg may be reliably administered intranasally; for doses >40 mg, part of the dose will be delivered to the oropharynx and ingested orally due to volume limitations, which may decrease effectiveness (Ref).

Chronic pain, intractable:

Note: Prior to use, consult with a pain specialist or service experienced with ketamine use in this setting. Optimal regimens, doses, and duration have not been identified; refer to institutional protocols. Long-term safety, efficacy, and timing of repeated ketamine treatments are not well established. Recommendations provided below are examples of regimens. May be used in refractory chronic pain (eg, complex regional pain syndrome, end-stage illness, neuropathies). May need to use doses at the higher range in opioid-tolerant patients (Ref). Reduce baseline opioids by 25% to 50% when used concomitantly with ketamine (Ref).

IV intermittent infusion: Initial: 0.25 to 0.6 mg/kg (usual maximum dose: 60 mg) as a 4- to 6-hour infusion; titrate to pain goal and tolerability; repeat daily for up to 2 to 10 days as needed (Ref).

IV continuous infusion: Initial: 0.05 to 0.15 mg/kg/hour (or 0.8 to 2.5 mcg/kg/minute) for 1 day outpatient or for 2 to 5 days inpatient; titrate to pain goal and tolerability; usual dosing range: 0.02 to 1 mg/kg/hour (or 0.3 to 16.7 mcg/kg/minute) (maximum dose: 30 mg/hour or 500 mcg/minute; not well established) (Ref).

SUBQ: Initial: 0.1 to 0.6 mg/kg (usual 2.5 to 25 mg) as needed; titrate to pain goal and tolerability. In patients who need a longer duration of analgesia, follow with continuous SubQ infusion at 0.1 to 1.2 mg/kg/hour (or 1.7 to 20 mcg/kg/minute) (maximum daily dose: 500 mg or 500,000 mcg) (Ref).

Oral: Note: Used in refractory chronic pain (eg, advanced illness or palliative care) in a hospitalized patient when other regimens have failed.

Initial: 0.5 mg/kg/day administered in 3 to 4 divided doses as needed; then increase dose in increments of ~5 mg/dose based on pain goal and tolerability; maximum daily escalation dose: 15 to 20 mg; maximum dose: 800 mg/day (Ref). **Note:** May administer dose as the undiluted injectable or mixed with an appropriate flavoring agent (eg, simple syrup).

Awake intubation, ICU or emergency room

Awake intubation, ICU or emergency room (off-label use):

Emergency room: IV: 25 to 50 mg every 5 to 10 minutes titrated to patient comfort and compliance (Ref).

Intensive care unit: IV: 10 to 20 mg every 5 to 10 minutes titrated to patient comfort and compliance (Ref).

Depressive episode associated with major depressive disorder

Depressive episode (severe, treatment resistant) associated with major depressive disorder (unipolar) (off-label use):

Note: IV route has been studied more than other routes (Ref). Ketamine may be used concurrently with an oral antidepressant and/or psychotherapy (Ref).

IV: Initial: 0.5 mg/kg administered over 40 minutes; may repeat at a frequency of 1 to 3 times weekly; may increase dose to 0.75 to 1 mg/kg based on response and tolerability. Treatment up to 6 weeks has been studied, although optimal duration of therapy is unknown (Ref).

IM: Initial: 0.25 to 0.5 mg/kg/dose (Ref). Although data are limited, additional doses have been described; however, dose and frequency are poorly defined (Ref).

General anesthesia

General anesthesia (alternative agent):

Induction of anesthesia:

Note: Useful in hypotensive patients or patients likely to develop hypotension during induction; lower doses may be used if concomitant anesthesia/sedatives (eg, midazolam) are administered (Ref).

IV: 0.5 to 2 mg/kg **or** 0.5 to 1 mg/kg in patients with shock (Ref).

IM (use only if IV access is not available): 4 to 6 mg/kg (Ref).

Maintenance of anesthesia (adjunct with total intravenous anesthesia): **Note:** Adjunct to reduce requirements of other anesthetic agents. Typically reserved for opioid-tolerant patients in order to avoid or limit opioid requirements (Ref); concurrent use of nitrous oxide reduces ketamine requirements (Ref).

IV: 0.25 to 0.35 mg/kg, followed by continuous infusion up to 1 mg/kg/hour (**or** 16.7 mcg/kg/minute) (Ref).

Mechanically ventilated patients in the ICU, analgesia/sedation/agitation

Mechanically ventilated patients in the ICU, analgesia/sedation/agitation (adjunct or alternative agent) (off-label use):

Note: Used as part of a multimodal strategy and adjunct for the reduction of opioid and sedative requirements and/or for the management of opioid-induced hyperalgesia in patients with acute/chronic pain or postsurgical pain. Pain should be monitored using validated scales (eg, behavioral pain scale, critical-care pain observation tool) in patients who are unable to self-report. If used for sedation, titrate to a light level of sedation (eg, Richmond Agitation Sedation Scale 0 to -2), unless deeper sedation is clinically indicated (Ref). Refer to institutional policies and procedures.

IV: Initial: 0.1 to 0.5 mg/kg bolus, followed by a 0.2 to 0.5 mg/kg/hour (**or** 3.3 to 8.3 mcg/kg/minute) continuous infusion; titrate dose to pain and/or sedation goal (dosing range: 0.04 to 2.5 mg/kg/hour **or** 0.67 to 41.7 mcg/kg/minute) (Ref). **Note:** Higher initial bolus doses (eg, 1 to 2 mg/kg) can be used to manage severely agitated patients (Ref).

Procedural sedation

Procedural sedation (off-label use): **Note:** Although rarely needed, may consider using midazolam to decrease risk **or** to treat emergence reactions. However, premedication may significantly prolong recovery time (Ref).

IV: Initial: 1 to 2 mg/kg over 1 to 2 minutes; if initial sedation is inadequate or for longer procedures, repeat dose (0.5 to 1 mg/kg) every 5 to 10 minutes; use lower doses (0.25 to 0.5 mg/kg) depending on concomitant sedation and clinical status (Ref). Some experts recommend to not exceed administering 2 mg/kg in a 30-minute period (Ref).

IM: **Note:** Onset of sedation will be delayed ~5 minutes with this route.

Initial: 4 to 5 mg/kg as a single dose; if sedation is inadequate after 5 to 10 minutes, repeat dose (2 to 5 mg/kg) (Ref).

Rapid sequence intubation outside the operating room, induction

Rapid sequence intubation outside the operating room, induction (off-label use):

IV: Initial: 1 to 2 mg/kg once over 1 minute (Ref). Some experts decrease dose in the setting of hemodynamic instability (eg, ~50% dose reduction **or** administer dose in 20 mg increments until patient is unconscious) (Ref).

Intraosseous (use only if IV access is not available): Initial: 100 mg once (Ref).

Status epilepticus, refractory

Status epilepticus, refractory (off-label use): Note: Used as an alternative or adjunct to midazolam, propofol, or barbiturates after conventional intermittent antiseizure therapies have failed. Mechanical ventilation and hemodynamic support generally required; continuous EEG is recommended; titrate doses to cessation of electrographic seizures or burst suppression (Ref). Optimal regimen and dose are uncertain; refer to institutional protocol.

IV:

Loading dose: Initial: 1.5 mg/kg **or** 0.5 to 3 mg/kg; repeat loading dose of 0.5 mg/kg every 3 to 5 minutes as needed for electrographic/burst suppression, followed by continuous infusion (Ref).

Continuous infusion: After initial loading dose and electrographic suppression, begin at a rate of 0.1 to 4 mg/kg/hour (**or** 1.7 to 66.7 **mcg/kg/minute**); titrate as needed for electrographic/burst suppression; maximum dose: 15 mg/kg/hour (**or** 250 **mcg/kg/minute**) (Ref).

Note: Generally, a period of at least 24 hours of electrographic suppression is suggested prior to down-titrating the continuous infusion; withdraw gradually by decreasing the dose by 20% every 3 hours while continuing conventional antiseizure therapies (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Note: Parenteral administration: Ketamine steady state concentrations are 20% higher in individuals with severe kidney dysfunction, corresponding to a moderate decrease in clearance. The dehydronorketamine metabolite, which may possess some degree of pharmacologic activity, accumulates in kidney dysfunction, although the clinical significance is unknown (Ref).

Oral administration (off-label route): Pharmacokinetic data regarding oral administration is very limited. When administered orally, norketamine (the primary active metabolite) and dehydronorketamine concentrations are higher compared to other routes of administration due to extensive first-pass metabolism in the liver; caution is warranted (Ref).

Altered kidney function:

Parenteral: No dosage adjustment necessary (Ref). Avoid prolonged use or monitor for signs of toxicity due to accumulation of dehydronorketamine (Ref).

Oral (off-label route): No initial dosage adjustment necessary. Use with caution and only with close monitoring in patients with severe kidney impairment (Ref).

Augmented renal clearance (measured urinary CrCl $\geq$ 130 mL/minute/1.73 m²):

Note: Augmented renal clearance (ARC) is a condition that occurs in certain critically ill patients without organ dysfunction and with normal serum creatinine concentrations. Younger patients (<55 years of age) admitted post trauma or major surgery are at highest risk for ARC, as well as those with sepsis, burns, or hematologic malignancies. An 8- to 24-hour measured urinary CrCl is necessary to identify these patients (Ref).

Parenteral, Oral (off-label route): No data available. No dosage adjustment necessary (Ref).

Hemodialysis, intermittent (thrice weekly): Unlikely to be significantly dialyzable (Ref).

Parenteral: No dosage adjustment necessary (Ref). Avoid prolonged use or monitor for signs of toxicity due to accumulation of dehydronorketamine (Ref).

Oral (off-label route): No initial dosage adjustment necessary. Use with caution and only with close monitoring (Ref).

Peritoneal dialysis: Unlikely to be significantly dialyzable (Ref).

Parenteral: No dosage adjustment necessary (Ref). Avoid prolonged use or monitor for signs of toxicity due to accumulation of dehydronorketamine (Ref).

Oral (off-label route): No initial dosage adjustment necessary. Use with caution and only with close monitoring (Ref).

CRRT: Removal by continuous hemodiafiltration not significant (Ref).

Parenteral: No dosage adjustment necessary (Ref). Avoid prolonged use or monitor for signs of toxicity due to accumulation of dehydronorketamine (Ref).

Oral (off-label route): No initial dosage adjustment necessary. Use with caution and only with close monitoring (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): No data available. Removal by PIRRT unlikely to be significant (Ref).

Parenteral: No dosage adjustment necessary (Ref). Avoid prolonged use or monitor for signs of accumulation or toxicity due to accumulation of dehydronorketamine (Ref).

Oral (off-label route): No initial dosage adjustment necessary. Use with caution and only with close monitoring (Ref).

Dosing: Liver Impairment: Adult

Liver impairment prior to treatment initiation:

Initial or dose adjustment in patients with preexisting liver cirrhosis: There are no dosage adjustments provided in the manufacturer's labeling.

Liver impairment developing in patients already receiving ketamine:

Ketamine-induced liver injury: If liver injury is suspected or confirmed, discontinue use immediately.

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1 and 2 obesity (BMI 30 to 39 kg/m²):

IV, IM, Intranasal, Oral, SUBQ: Use **actual body weight** for weight-based dosing, then titrate to clinical effect, if needed (Ref). Refer to adult dosing for indication-specific doses.

Class 3 obesity (BMI ≥40 kg/m²):

IV, IM, Intranasal, Oral, SUBQ: Use either **adjusted body weight** or **ideal body weight** for initial weight-based dose calculations, then titrate to clinical effect (Ref). If rapid sedation or clinical effect is needed, use **adjusted body weight** for initial weight-based dose calculations, then titrate to clinical effect (Ref). Clinicians should **not** change dosing weight from one weight metric to another during therapy (ie, actual body weight to/from adjusted body weight or ideal body weight) (Ref). Refer to adult dosing for indication-specific doses.

Rationale for recommendations: Ketamine, a short-acting IV induction anesthetic, has a large V_d (2.1 to 3.1 L/kg), with peak anesthetic effects occurring within minutes and a rapid loss of effect related to redistribution (Ref). There are limited data on the influence of obesity on ketamine dosing or pharmacokinetics. A variety of studies have been performed in the past decade for acute pain and depression in patients with obesity or severe obesity using lean body weight, ideal body weight, and actual body weight; however, no consistency between dosing weight and clinical effect has been demonstrated (Ref). This inconsistency (even within indication) in specific weights used for dosing limits definition of a universal weight-based dosing strategy. Due to ketamine's active metabolite, prolonged use may contribute to the risk of accumulation and toxicity, especially in patients with BMI ≥ 40 kg/m² (Ref).

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Ketamine: Pediatric drug information")

Dosage guidance:

Safety: The American College of Emergency Physicians considers the use of ketamine in infants <3 months of age to be an absolute contraindication, due to the higher risk of airway complications (Ref).

Dosing: Individualize dose and titrate to effect. May be used in combination with anticholinergic agents to decrease hypersalivation.

Acute pain

Acute pain (low dose; sub-dissociative): Very limited data available: Optimal dose not established: **Note:** Use of mucosal atomizer device is recommended:

Children ≥ 3 years and Adolescents: Intranasal: Usual: 1 mg/kg/dose, may repeat once; range: 0.5 to 1.5 mg/kg/dose; maximum dose: 100 mg/dose (Ref). Dosing based on 2 double-blind, randomized, controlled studies (n=80 in ketamine treatment group, age range: 3 to 17 years) comparing intranasal ketamine to intranasal fentanyl in the treatment of acute pain due to isolated limb injury or suspected single extremity fracture in the emergency department; pain relief was similar between both treatment groups; no serious adverse effects were reported (Ref).

Anesthesia

Anesthesia:

Pre-anesthetic sedation: Limited data available:

Intranasal:

Infants ≥ 6 months: 3 mg/kg/dose (half dose per nostril) administered at least 15 minutes prior to mask induction (Ref).

Children <2 years: 3 to 5 mg/kg/dose (half dose per nostril) administered at least 15 minutes prior to mask induction (Ref).

Children 2 to 7 years: 3 to 6 mg/kg/dose (half dose per nostril) administered 15 to 40 minutes prior to induction (Ref).

Oral: Children ≤ 8 years: 6 to 8 mg/kg/dose 20 to 30 minutes prior to surgery. Dosing based on 2 prospective, randomized, double-blind, placebo-controlled, dose finding trials. The larger trial compared 4 mg/kg, 6 mg/kg, and 8 mg/kg (n=20 in each group; age range: 2 to 8 years) and found that patients who received 8 mg/kg were significantly more calm, but also had slightly longer recovery times. Sedation was effective within 10 minutes of administration in 80% and 45% of the patients in the 8 mg/kg and 6 mg/kg, groups respectively, but was not effective in the 4 mg/kg group (Ref). The other study compared 3 mg/kg and 6 mg/kg (n=15 in each group, age range: 1 to 7 years) and found that patients who received 6 mg/kg had satisfactory sedation without prolonged recovery

times, but the 3 mg/kg dose did not provide uniform sedation nor offer a significant improvement in premedicated emotional state compared to placebo (Ref).

Rectal: Administer 15 to 45 minutes prior to surgery as a single agent; when used in combination with other sedatives, lower doses should be considered. Efficacy was reported in trials comparing rectal ketamine to rectal doses of other agents (fentanyl/droperidol, midazolam) (Ref). Reported effective range:

Infants 2 to 6 months: 8 mg/kg/dose.

Infants ≥ 7 months and Children ≤ 9 years: 8 to 10 mg/kg/dose.

Note: Although lower doses of 4 to 7 mg/kg/dose have been reported, they were less effective. In some patients, the 10 mg/kg/dose was associated with prolonged postoperative sedation. When used in combination with midazolam, a lower rectal dose of 3 mg/kg/dose has been effective (Ref).

Induction of anesthesia:

Infants ≥ 3 months, Children, and Adolescents < 16 years: Limited data available:

IM: 5 to 10 mg/kg has been reported and suggested by experts (Ref).

IV: 1 to 3 mg/kg has been reported and suggested by experts (Ref).

Adolescents ≥ 16 years:

IM: 6.5 to 13 mg/kg.

IV: 1 to 4.5 mg/kg.

Maintenance of anesthesia: Adolescents ≥ 16 years: May administer supplemental doses of one-half to the full induction dose as needed.

Endotracheal intubation

Endotracheal intubation: Limited data available: Infants, Children, and Adolescents: IV: 1 to 2 mg/kg as part of rapid sequence sedation (Ref).

Sedation/Analgesia, procedural

Sedation/Analgesia, procedural: Limited data available:

Ketamine without propofol: Infants ≥ 3 months, Children, and Adolescents:

IM: 4 to 5 mg/kg as a single dose; may give a repeat dose (range: 2 to 5 mg/kg) if sedation inadequate after 5 to 10 minutes or if additional doses are required (Ref). Some have recommended smaller doses (2 to 2.5 mg/kg) for minor procedures (eg, wound suture with local anesthetic) (Ref).

IV: 1 to 2 mg/kg over 30 to 60 seconds. If initial sedation inadequate or repeated doses are necessary to accomplish a longer procedure, may administer additional doses of 0.5 to 1 mg/kg every 5 to 15 minutes as needed (Ref).

Intranasal: **Note:** Use of mucosal atomizer device is recommended:

Infants ≥ 3 months and Children: 3 to 6 mg/kg (half dose per nostril), doses up to 9 mg/kg have been described (Ref).

Oral: Children and Adolescents: 5 mg/kg with oral midazolam given 30 to 45 minutes before the procedure. Dosing based on 2 prospective, randomized, blinded studies involving patients aged 1 to 10 years with either laceration repair or burn wound care (Ref). A lower dose (3 mg/kg) in addition to midazolam was effective in a study comparing different routes (IV, oral, rectal) of ketamine plus midazolam for invasive procedures in oncology patients. The oral group included 24 patients (mean age: 3.9 ± 1.3 years); incidence of optimal sedation was similar between groups (75% for the oral group) (Ref). **Note:** A higher dose (10 mg/kg) has been used successfully as monotherapy prior to procedures in pediatric oncology patients (n=35, age: 14 months to 17 years; mean age: 6.5 years) (Ref).

Rectal: Children 1 to 8 years: 1.5 to 3 mg/kg with midazolam as a single dose 20 minutes prior to painful procedure. Dosing based on 2 studies. The first was completed in children with burns requiring dressing changes (n=47 procedures in 30 patients, mean age: 1.9 years, range: 10 months to 7.3 years). Patients received 0.75 mg/kg of the S(+) isomer (equivalent to 1.5 mg/kg of racemic ketamine) along with rectal midazolam 20 minutes prior to dressing changes; 94% of the procedures were reported to have good or excellent analgesia (Ref). The second study compared different routes (IV, oral, rectal) of ketamine plus midazolam for invasive procedures in oncology patients. The rectal group included 24 patients (mean age: 3.7 ± 1.1 years) who received 3 mg/kg of ketamine in addition to midazolam. Incidence of optimal sedation was similar between groups (79% for the rectal group) (Ref). In both of these studies, rectal ketamine was well tolerated.

Ketamine with propofol ("ketofol"): Infants ≥ 3 months, Children, and Adolescents: IV: 0.5 to 0.75 mg/kg of each agent. This combination has been used to decrease the dose of each agent required. It has been proposed that these lower doses help decrease adverse effects, ketamine may decrease the propofol-related hypotension and respiratory depression, and propofol may decrease the ketamine associated nausea and emergence reactions (Ref).

Sedation/Analgesia, critically ill patients

Sedation/Analgesia, critically ill patients: Very limited data available: Infants ≥ 5 months, Children, and Adolescents: Initial dose: IV: 0.5 to 2 mg/kg, then continuous IV infusion: 5 to 20 mcg/kg/minute (0.3 to 1.2 **mg/kg/hour**); start at lower dosage listed and titrate to effect (Ref); doses as high as 60 mcg/kg/minute (3.6 **mg/kg/hour**) have been reported in patients with refractory bronchospasm (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions (Significant): Considerations

Airway complications

When used for procedural sedation for major procedures involving the posterior pharynx (eg, endoscopy) or in patients with an active pulmonary infection or disease (including upper respiratory disease or asthma), the use of ketamine increases the risk of **laryngospasm** (rare and typically transient) (Ref). The manufacturer recommends against the use of ketamine alone in surgery or diagnostic procedures of the pharynx, larynx, or bronchial tree; mechanical stimulation of the pharynx should be avoided, whenever possible, if ketamine is used alone.

Mechanism: Dose-related; ketamine does not suppress pharyngeal and laryngeal reflexes. Laryngospasm may occur secondary to stimulation of the vocal cords by instrumentation and/or ketamine-induced increased salivary secretions (Ref).

Risk factors:

- Higher IV doses (Ref)
- IM administration (and higher doses) (Ref)
- Pediatric patients < 2 years or ≥ 13 years of age (Ref)
- Active pulmonary infection or disease, including upper respiratory disease or asthma (Ref)
- History of airway instability, tracheal surgery, or tracheal stenosis (Ref)

Cardiovascular effects

Ketamine causes **increased blood pressure, increased heart rate**, and increased cardiac output, thereby increasing myocardial oxygen demand; **cardiac arrhythmia**, cardiac decompensation, **decreased blood pressure**, and **decreased heart rate** have also been reported (Ref). In a scientific statement from the American Heart Association, ketamine has been determined to be an agent that may exacerbate underlying myocardial dysfunction (magnitude: major) (Ref). Resolution of increased blood pressure usually occurs within 15 minutes after the peak.

Mechanism: Ketamine has both central sympathetic stimulation and inhibition of catecholamine uptake and direct negative inotropic effects. Central sympathetic stimulation results in increased blood pressure, heart rate, cardiac output, and therefore, increased myocardial oxygen demand (Ref). Direct negative inotropic effects may predominate in patients with heart failure and contribute to decompensation (Ref).

Onset: Rapid; a peak in blood pressure occurs within a few minutes of administration.

Risk factors:

- Known or suspected coronary artery disease, angina, hypertension (Ref)
- Older adults with risk factors for coronary artery disease (Ref)
- Heart failure (Ref)
- Catecholamine depletion (hypotension) (Ref)

Dependence

Ketamine may cause **drug dependence** (withdrawal symptoms on discontinuation) and **drug tolerance** with prolonged use (Ref). Multiple case reports of ketamine misuse have been described (Ref). A **withdrawal syndrome** following discontinuation may include symptoms such as shaking, sweating, palpitations, fatigue, decreased appetite, chills, autonomic arousal, lacrimation, restlessness, cravings, dysphoria, anxiety, depressed mood, nightmares, paranoia, delusions, and hallucinations (Ref). Typical duration of withdrawal symptoms is ~3 days, with some cases persisting for 2 weeks (Ref).

Mechanism: Dose- and time-related; development of dependence may be related to the psychological effects of ketamine and tolerance. Ketamine is structurally similar to phencyclidine and the psychological effects contributing to dependence may include euphoria, perceptual changes, dissociation, and hallucinations. Tolerance most likely occurs due to auto-induction of metabolism (Ref). Dependence may also be related to N-methyl-D-aspartate (NMDA) glutamate receptor blockade, producing alcohol-like subjective effects, as well as opioid effects (Ref).

Onset: Dependence: Delayed; from isolated case reports, dependence may evolve over weeks to years (Ref). Withdrawal syndrome: Rapid; symptoms usually occur within 24 hours of discontinuation (Ref).

Risk factors:

- Long-term high-dose use (Ref)
- Rapid dose escalation (Ref)
- History of multiple substance use disorders (eg, alcohol, cannabis, opioids) (Ref)
- Ease of access (Ref)

Emergence reactions

Prolonged emergence from anesthesia, which can manifest as vivid dreams, **hallucinations**, and/or frank **delirium** commonly occur in all ages (Ref).

Mechanism: Dose-related; may occur due to depression of auditory and visual relay nuclei, leading to misperception and/or misinterpretation of auditory and visual stimuli (Ref).

Onset: Rapid; emergence reactions may occur up to 24 hours postoperatively.

Risk factors:

- Higher doses (Ref)
- Rapid IV administration (Ref)
- Age >16 years (Ref)
- Females (Ref)
- Excessive noise or stimulation during recovery (Ref)
- People who normally dream or have a history of a personality disorder (Ref)
- Ketamine monotherapy (concurrent use with a benzodiazepine may reduce risk) (Ref)

Genitourinary effects

Lower urinary tract and bladder symptoms, including **dysuria, urinary frequency, urinary urgency, urinary incontinence, hematuria, and nocturia**, have been reported in patients with a history of chronic ketamine use or abuse and may be related to ketamine treatment, not the underlying condition. Additional findings from diagnostic studies have included **cystitis, hydronephrosis, and bladder dysfunction** (reduced capacity) (Ref). Reversibility of symptoms depends on time to diagnosis and intervention (Ref).

Mechanism: Not clearly established; several proposed mechanisms, including direct effects of ketamine and the active metabolite, norketamine/hydroxynorketamine on bladder mucosa (Ref).

Onset: Delayed; case reports range from 1 month to 4.5 years (Ref).

Risk factors:

- Chronic use or abuse (Ref)
- Frequency of use of other drugs (eg, opiates, methamphetamine) (Ref)
- Female gender (Ref)
- History of urinary tract infections (Ref)

Neurodevelopmental effects in children

Animal studies have shown that prolonged or repeated exposure to medications for anesthesia or sedation cause adverse effects on brain maturation resulting in changes in behavior and learning. Some human studies have also suggested similar effects, including epidemiological studies in humans that have observed various cognitive and behavioral problems, including neurodevelopmental delay (and related diagnoses), learning disabilities, and attention deficit hyperactivity disorder. However, data are limited, inconclusive, and further studies are needed to fully characterize findings. Based on the potential risk, the FDA warned in 2016 that in neonates, infants, children <3 years, and patients in the third trimester of pregnancy (ie, times of rapid brain growth and synaptogenesis) undergoing repeated or lengthy exposure to sedatives or anesthetics during surgery/procedures/critical illness, there may be detrimental effects on the child's or fetus' brain development which may lead to various cognitive and behavioral problems. Relatively short exposure (<3 hours) to sedatives or anesthetics during surgery or procedures is unlikely to adversely affect brain development (Ref).

Mechanism: Unknown; in juvenile animal studies, drugs that potentiate gamma-aminobutyric acid activity and/or block N-methyl-D aspartate receptors for >3 hours demonstrated widespread neuronal and oligodendrocyte cell loss along with alteration in synaptic morphology and neurogenesis (Ref).

Risk factors:

- Neonates, infants, children <3 years, and pregnant patients in their third trimester undergoing procedures lasting >3 hours or multiple procedures (Ref)

Respiratory depression

Ketamine may rarely cause **respiratory depression** or **apnea**, which are usually transient. These adverse reactions are most associated with rapid IV administration but have also been reported

with IM administration (Ref). According to the manufacturer, overdose is also a risk factor for these adverse reactions; however, respiratory depression and apnea have been reported in the setting of slow sub-anesthetic IV infusion (Ref).

Onset: Rapid; typically, 1 to 2 minutes after IV administration or 4 to 5 minutes after IM administration (Ref). In the case of respiratory depression and apnea in the setting of slow sub-clinical IV infusion, onset occurred 25 minutes after the start of ketamine infusion (Ref).

Risk factors:

- Overdose
- Rapid IV administration (Ref)
- Concurrent use of other CNS depressants (eg, benzodiazepines, opioids)

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%: Nervous system: Prolonged emergence from anesthesia (12%; including agitation, confusion, delirium, dreamlike state, excitement, hallucinations, irrational behavior, vivid imagery)

Frequency not defined:

Cardiovascular: Cardiac arrhythmia, cardiac decompensation, decreased blood pressure, decreased heart rate, increased blood pressure, increased heart rate

Dermatologic: Erythema of skin, morbilliform rash

Gastrointestinal: Anorexia, nausea, vomiting

Local: Pain at injection site, rash at injection site

Nervous system: Hypertonia (tonic-clonic movements sometimes resembling seizures), psychiatric disturbance

Ophthalmic: Diplopia, increased intraocular pressure, nystagmus disorder

Respiratory: Airway obstruction

Postmarketing:

Cardiovascular: Premature ventricular contractions (Ref)

Endocrine & metabolic: Arginine vasopressin deficiency (Ref)

Gastrointestinal: Cholangitis (Ref), sclerosing cholangitis, sialorrhea (Ref)

Genitourinary: Bladder dysfunction (reduced capacity) (Ref), cystitis (including cystitis noninfective, cystitis interstitial, cystitis ulcerative, cystitis erosive, cystitis hemorrhagic) (Ref), dysuria (Ref), hematuria (Ref), nocturia (Ref), urethral obstruction, urethral stricture, urinary frequency (Ref), urinary incontinence (Ref), urinary urgency (Ref)

Hepatic: Abnormal hepatobiliary function (with recurrent or continuous use; including biliary duct dilation, biliary duct stricture, biliary obstruction, biliary tract stenosis, pericholeductal fibrosis) (Ref), hepatic cirrhosis (Ref), increased direct serum bilirubin (Ref), increased gamma-glutamyl transferase (Ref), increased liver enzymes (Ref), increased serum alanine aminotransferase (Ref), increased serum alkaline phosphatase (Ref), increased serum aspartate aminotransferase (Ref), increased serum bilirubin (Ref), jaundice (Ref)

Hypersensitivity: Anaphylaxis (Ref)

Nervous system: Drug dependence (Ref), increased cerebrospinal fluid pressure (Ref), withdrawal syndrome (Ref)

Neuromuscular & skeletal: Laryngospasm (Ref)

Renal: Hydronephrosis (Ref)

Respiratory: Apnea (Ref), respiratory depression (Ref)

Miscellaneous: Drug tolerance

Contraindications

Hypersensitivity to ketamine or any component of the formulation; conditions in which an increase in blood pressure would be hazardous

Note: When used for procedural sedation and analgesia in the emergency department, the following additional absolute contraindications according to the American College of Emergency Physicians have been asserted (ACEP [Green 2011]): Infants <3 months of age; known or suspected schizophrenia (even if currently stable or controlled with medications)

Canadian labeling: Additional contraindications (not in US labeling): History of cerebrovascular accident; severe cardiac decompensation; surgery of the pharynx, larynx, or bronchial tree unless adequate muscle relaxants are used

Warnings/Precautions

Concerns related to adverse effects:

- **CNS depression:** May cause CNS depression, which may impair physical or mental abilities; patients must be cautioned about performing tasks that require mental alertness (eg, operating machinery, driving). When used for outpatient surgery, the patient should be accompanied by a responsible adult. Driving, operating hazardous machinery, or engaging in hazardous activities should not be undertaken for ≥ 24 hours after anesthesia, according to the manufacturer.
- **Increased intracranial pressure:** Some consider the use of ketamine in patients with CNS masses, CNS abnormalities, or hydrocephalus a relative contraindication due to multiple reports that ketamine may increase intracranial pressure in these patients; use caution, especially at higher doses (ACEP [Green 2011]; Cohen 2018). However, assuming adequate ventilation, some evidence suggests that ketamine has minimal effects on intracranial pressure and may even improve cerebral perfusion and reduce intracranial pressure (Albanese 1997; Bowles 2012; Quibell 2015; Zeiler 2014).
- **Increased ocular pressure:** Use with caution in patients with increased intraocular pressure (IOP). Some recommend avoiding use in patients with an open eye injury or other ophthalmologic disorder where an increase in IOP would prove to be detrimental; however, the effects of ketamine on IOP is mixed with some evidence demonstrating no clinically significant effect on IOP (ACEP [Green 2011]; Cunningham 1986; Drayna 2012; Miller 2010; Nagdeve 2006).
- **Liver injury:** Recurrent use (eg, abuse/misuse, medically supervised unapproved use) may cause hepatobiliary dysfunction (most often a cholestatic pattern); biliary duct dilatation, stricture, stenosis, and obstructions have also been reported with recurrent use. Sclerosing cholangitis (eg, cholestatic pattern of increased LFTs with grossly increased gamma glutamyl transferase and alkaline phosphatase levels) has been reported in patients on long-term therapy; potentially reversible with ketamine discontinuation.
- **Porphyria:** The ACEP considers the use of ketamine in patients with porphyria a relative contraindication due to enhanced sympathomimetic effect produced by ketamine (ACEP [Green 2011]).
- **Thyroid disorders:** The ACEP considers the use of ketamine in patients with a thyroid disorder or receiving a thyroid medication a relative contraindication due to enhanced sympathomimetic effect produced by ketamine (ACEP [Green 2011]).

Disease-related concerns:

- **Cerebrospinal fluid pressure elevation:** Use with caution in patients with cerebrospinal fluid (CSF) pressure elevation; an increase in CSF pressure may be associated with use.
- **Ethanol use:** Use with caution in patients with chronic alcohol use disorder or who are acutely alcohol-intoxicated..

Special populations:

- Pediatric neurotoxicity: In pediatric and neonatal patients <3 years of age and patients in third trimester of pregnancy (ie, times of rapid brain growth and synaptogenesis), the repeated or lengthy exposure to sedatives or anesthetics during surgery/procedures may have detrimental effects on child or fetal brain development and may contribute to various cognitive and behavioral problems. Epidemiological studies in humans have reported various cognitive and behavioral problems, including neurodevelopmental delay (and related diagnoses), learning disabilities, and attention deficit hyperactivity disorder. Human clinical data suggest that single, relatively short exposures are not likely to have similar negative effects. No specific anesthetic/sedative has been found to be safer. For elective procedures, risk versus benefits should be evaluated and discussed with parents/caregivers/patients; critical surgeries should not be delayed (FDA 2016).

Other warnings/precautions:

- Experienced personnel: Use requires careful patient monitoring, should only be used by experienced personnel who are not actively engaged in the procedure or surgery. If used in a nonintubated and/or nonmechanically ventilated patient, qualified personnel and appropriate equipment for rapid institution of respiratory and/or cardiovascular support must be immediately available. Use to induce moderate (conscious) sedation in patients warrants monitoring equivalent to that seen with deep anesthesia. Consult local regulations and individual institutional policies and procedures.

Warnings: Additional Pediatric Considerations

Postanesthetic emergence reactions may be minimized by limiting verbal, tactile, and visual patient stimulation during recovery, or by pretreatment with a benzodiazepine (using lower recommended doses of ketamine). Severe emergent reactions may require treatment with a small hypnotic dose of a short or ultrashort acting barbiturate.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Injection, as hydrochloride:

Ketalar: 10 mg/mL (20 mL); 50 mg/mL (10 mL); 100 mg/mL (5 mL)

Generic: 10 mg/mL (20 mL); 50 mg/mL (10 mL); 100 mg/mL (5 mL, 10 mL [DSC])

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Ketalar Injection)

10 mg/mL (per mL): \$1.19

50 mg/mL (per mL): \$0.84

100 mg/mL (per mL): \$3.20

Solution (Ketamine HCl Injection)

10 mg/mL (per mL): \$1.19 - \$1.46

50 mg/mL (per mL): \$0.43 - \$1.19

100 mg/mL (per mL): \$1.33 - \$3.62

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price

for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection, as hydrochloride:

Ketalar: 10 mg/mL (20 mL); 50 mg/mL (10 mL, 20 mL) [contains benzethonium chloride]

Generic: 10 mg/mL (2 mL, 20 mL); 50 mg/mL (2 mL, 10 mL)

Controlled Substance

C-III

Administration: Adult

IM: Inject deep IM into large muscle mass.

Intranasal (off label): Administer half dose in each nostril using a needleless syringe or mucosal atomizer device. **Note:** Intranasal use for adults for procedural sedation is not recommended since volume limits an adequate dose (Ref).

IV: According to the manufacturer, administer bolus/induction doses over 1 minute or at a rate of 0.5 mg/kg/minute; more rapid administration may result in respiratory depression and enhanced pressor response. Some experts suggest administration over 2 to 3 minutes (Ref). When used for treatment refractory unipolar depression, administer over 40 minutes (Ref). May also be administered as a continuous infusion.

Oral (off label): May administer undiluted or mix the appropriate dose (using the injectable solution) in a flavoring agent (eg, simple syrup), cola, or other beverage; administer immediately after preparation (Ref).

Rectal (off label): May use the 50 mg/mL solution undiluted or use the 100 mg/mL solution and further dilute prior to administration (Ref).

SUBQ (off label): May administer intermittent doses SUBQ. May also administer as a SUBQ continuous infusion using a prepared dilution (eg, 200 mg in 50 mL NS) (Ref).

Preparation for Administration: Adult

Intranasal (off label): Use the 50 or 100 mg/mL injectable solution; may administer undiluted or further diluted in NS to a concentration of 20 mg/mL (Ref).

IM: The 100 mg/mL concentration may be administered without dilution for induction of anesthesia.

IV:

IV bolus: Do not administer the 100 mg/mL concentration as IV bolus unless first properly diluted with at least an equal volume of SWFI, NS, or D5W. The 10 mg/mL and 50 mg/mL concentrations may be administered without dilution for induction of anesthesia.

IV continuous infusion: Further dilute the 50 mg/mL or 100 mg/mL vials in D5W or NS to prepare a maintenance infusion with a final concentration of 1 mg/mL (or 2 mg/mL in patients with fluid restriction); mix well. The 10 mg/mL vials may be administered without dilution for maintenance of anesthesia. Do not mix with barbiturates or diazepam (precipitation may occur).

Oral solution (off label): Use the 100 mg/mL injectable solution (10 mL vial); may administer undiluted or further dilute with simple syrup, purified water (90 mL), or may add own flavoring agent (eg, fruit cordial) to prepare 100 mL concentration of 50 mg per 5 mL; dilute immediately before administration (Ref).

SUBQ continuous infusion (off label): Using the 100 mg/mL concentration, further dilute 200 mg in 50 mL of NS to prepare a solution with a final concentration of 4 mg/mL (Ref).

Usual Infusion Concentrations: Adult

IV infusion: 1 mg/mL, 2 mg/mL, **or** 10 mg /mL.

Administration: Pediatric

Intranasal (using parenteral dosage form): Use the 50 or 100 mg/mL solution; may administer undiluted or further diluted in NS to a concentration of 20 mg/mL (Ref). Administer dose in 1 nostril (for volumes ≤ 0.5 mL) or divide dose and administer half dose in each nostril using a mucosal atomizer device (preferred) or needleless syringe (Ref).

Oral (using parenteral dosage form): Use the 100 mg/mL IV solution and mix the appropriate dose in 0.2 to 0.4 mL/kg of cola, sour cherry juice, or other beverage (Ref).

Parenteral:

IM: Administer into deep muscle mass; in a pediatric study, administration into lateral thigh is described (Ref).

IV push: May administer the 10 mg/mL and 50 mg/mL undiluted. Administer slowly over 60 seconds, do not exceed 0.5 mg/kg/minute; more rapid administration may result in respiratory depression and enhanced pressor response. Some experts suggest administration over 2 to 3 minutes (Ref).

IV infusion: May be administered as a continuous IV infusion.

Rectal (using parenteral dosage form): May use the 50 mg/mL solution undiluted or use the 100 mg/mL solution and further dilute prior to administration (Ref).

Preparation for Administration: Pediatric

Note: Three concentrations are available (10 mg/mL, 50 mg/mL, and 100 mg/mL); preparation information may be product specific.

Intranasal (using parenteral dosage form): Use the 50 mg/mL or 100 mg/mL solution; may administer undiluted or further dilute in NS to a concentration of 20 mg/mL (Ref).

Oral (using parenteral dosage form): Use the 100 mg/mL solution and mix the appropriate dose in 0.2 to 0.4 mL/kg of cola, sour cherry juice, or other beverage (Ref).

Parenteral:

IV push: If using the 100 mg/mL concentration, must further dilute with an equal volume of SWFI, NS, or D5W; maximum concentration: 50 mg/mL.

IV infusion: The 50 mg/mL and 100 mg/mL vials may be further diluted in D5W or NS to a final concentration of 1 mg/mL; in fluid restricted patients a concentration of 2 mg/mL may be used. ASHP recommends standard concentrations of 2 mg/mL and 10 mg/mL (undiluted) for continuous infusions in pediatric patients (Ref).

Note: Do not mix with barbiturates or diazepam (precipitation may occur).

Storage/Stability

Vials: Store intact vials at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Protect from light.

Oral solution (off label): Store prepared oral solution at 2°C to 8°C (36°F to 46°F) for up to 1 week from preparation (Quibell 2015).

Use: Labeled Indications

Anesthesia: Induction and maintenance of general anesthesia.

Use: Off-Label: Adult

Agitation, acute/severe or refractory; Analgesia, subanesthetic dosing, acute and chronic pain; Awake intubation, ICU or emergency room (off-label use); Depressive episode (severe, treatment resistant) associated with major depressive disorder (unipolar); Mechanically ventilated patients in the ICU, analgesia/sedation/agitation; Procedural sedation; Rapid sequence intubation outside the operating room, induction; Status epilepticus, refractory

Medication Safety Issues

Sound-alike/look-alike issues:

Ketalar may be confused with Kenalog, ketorolac

Ketamine may be confused with ketorolac

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (anesthetic agent, general, inhaled and IV; moderate and minimal sedation agent, oral, for children [using parenteral form]) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024; ISMP List of High-Alert Medications in Community/Ambulatory Care Settings ©ISMP 2021).

Older Adult: High-Risk Medication:

Ketamine is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥ 65 years of age) for the treatment of depression due to an increased risk of falls (O'Mahony 2023).

Metabolism/Transport Effects

Substrate of CYP2B6 (Major), CYP2C9 (Minor), CYP3A4 (Major with inducers), CYP3A4 (Minor with inhibitors); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Acrivastine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Alcohol (Ethyl): CNS Depressants may increase CNS depressant effects of Alcohol (Ethyl). *Risk C: Monitor*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amitriptyline: CNS Depressants may increase CNS depressant effects of Amitriptyline. *Risk C: Monitor*

Amitriptylinoxide: CNS Depressants may increase CNS depressant effects of Amitriptylinoxide. *Risk C: Monitor*

Amoxapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Asenapine: CNS Depressants may increase CNS depressant effects of Asenapine. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Baclofen: CNS Depressants may increase CNS depressant effects of Baclofen. *Risk C: Monitor*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Blonanserin: CNS Depressants may increase CNS depressant effects of Blonanserin. Management: Use caution if coadministering blonanserin and CNS depressants; dose reduction of the other CNS depressant may be required. Strong CNS depressants should not be coadministered with blonanserin. *Risk D: Consider Therapy Modification*

Brexanolone: CNS Depressants may increase CNS depressant effects of Brexanolone. *Risk C: Monitor*

Brimonidine (Ophthalmic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brivaracetam: CNS Depressants may increase CNS depressant effects of Brivaracetam. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brompheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bucizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

CarBAMazepine: CNS depressant effects of CarBAMazepine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Carisoprodol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Cenobamate: CNS depressant effects of Cenobamate and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cetirizine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk D: Consider Therapy Modification*

Chloral Hydrate-Chloral Betaine: CNS Depressants may increase CNS depressant effects of Chloral Hydrate-Chloral Betaine. Management: Consider alternatives to the use of chloral hydrate or chloral betaine and additional CNS depressants. If combined, consider a dose reduction of either agent and monitor closely for enhanced CNS depressive effects. *Risk D: Consider Therapy Modification*

Chlormethiazole: May increase CNS depressant effects of CNS Depressants. Management: Monitor closely for evidence of excessive CNS depression. The chlormethiazole labeling states that an appropriately reduced dose should be used if such a combination must be used. *Risk D: Consider Therapy Modification*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Chlorpheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: CNS depressant effects of ChlorproMAZINE and CNS Depressants may increase with coadministration. Management: If possible, discontinue the use of other CNS depressants prior to initiating chlorpromazine; other CNS depressants can be re-started at low doses and titrated slowly as needed. Monitor for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Chlorprothixene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Chlorzoxazone: CNS depressant effects of Chlorzoxazone and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cinnarizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clemastine: CNS depressant effects of Clemastine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

ClomiPRAMINE: CNS depressant effects of ClomiPRAMINE and CNS Depressants may increase with coadministration. *Risk C: Monitor*

CloNIDine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clothiapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

CloZAPine: CNS Depressants may increase CNS depressant effects of CloZAPine. *Risk C: Monitor*

CNS Depressants: May increase CNS depressant effects of Ketamine. *Risk C: Monitor*

Cyclizine: CNS depressant effects of Cyclizine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cyclobenzaprine: CNS depressant effects of CNS Depressants and Cyclobenzaprine may increase with coadministration. *Risk C: Monitor*

CYP2B6 Inducers (Moderate): May decrease serum concentrations of Ketamine. *Risk C: Monitor*

CYP3A4 Inducers (Moderate): May decrease serum concentrations of Ketamine. *Risk C: Monitor*

CYP3A4 Inducers (Strong): May decrease serum concentrations of Ketamine. *Risk C: Monitor*

CYP3A4 Inhibitors (Strong): May increase serum concentrations of Ketamine. *Risk C: Monitor*

Cyproheptadine: CNS depressant effects of Cyproheptadine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Dantrolene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Daridorexant: May increase CNS depressant effects of CNS Depressants. Management: Dose reduction of daridorexant and/or any other CNS depressant may be necessary. Use of daridorexant with alcohol is not recommended, and the use of daridorexant with any other drug to treat insomnia is not recommended. *Risk D: Consider Therapy Modification*

Desipramine: CNS Depressants may increase CNS depressant effects of Desipramine. *Risk C: Monitor*

Deutetrabenazine: CNS depressant effects of CNS Depressants and Deutetrabenazine may increase with coadministration. *Risk C: Monitor*

Dexbrompheniramine: CNS Depressants may increase CNS depressant effects of Dexbrompheniramine. *Risk C: Monitor*

Dexchlorpheniramine: CNS depressant effects of CNS Depressants and Dexchlorpheniramine may increase with coadministration. *Risk C: Monitor*

Dexmedetomidine: CNS Depressants may increase CNS depressant effects of Dexmedetomidine. Management: Monitor for increased CNS depression during coadministration of dexmedetomidine and CNS depressants, and consider dose reductions of either agent to avoid excessive CNS depression. *Risk D: Consider Therapy Modification*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Difenoxin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimethindene (Systemic): CNS Depressants may increase CNS depressant effects of Dimethindene (Systemic). *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Diphenhydramine (Systemic): CNS depressant effects of CNS Depressants and Diphenhydramine (Systemic) may increase with coadministration. *Risk C: Monitor*

Diphenoxylate: May increase CNS depressant effects of CNS Depressants. Management: Avoid coadministration of diphenoxylate/atropine with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients closely for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Dothiepin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Doxepin (Systemic): CNS depressant effects of CNS Depressants and Doxepin (Systemic) may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Doxepin (Systemic). Specifically, the risk of abnormal thinking or performing complex behaviors while asleep may be increased. Management: Monitor patients for CNS depression (eg, sedation, somnolence) if doxepin is coadministered with CNS depressants. If patients experience abnormal thinking or perform complex behaviors while asleep (eg, driving) strongly consider discontinuing doxepin. *Risk C: Monitor*

Doxepin (Topical): CNS Depressants may increase CNS depressant effects of Doxepin (Topical). *Risk C: Monitor*

Doxylamine: CNS Depressants may increase CNS depressant effects of Doxylamine. *Risk C: Monitor*

Droperidol: May increase CNS depressant effects of CNS Depressants. Management: Consider dose reductions of droperidol or of other CNS agents (eg, opioids, barbiturates) with concomitant use. *Risk D: Consider Therapy Modification*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Esketamine (Injection): CNS Depressants may increase therapeutic effects of Esketamine (Injection). Specifically, duration of esketamine injection may be prolonged. CNS Depressants may decrease adverse/toxic effects of Esketamine (Injection). *Risk C: Monitor*

Esketamine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Eszopiclone: CNS depressant effects of CNS Depressants and Eszopiclone may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Eszopiclone. Specifically, the risk of daytime psychomotor impairment may be increased. Management: Consider reducing the dose of eszopiclone or other CNS depressants if these agents are used concomitantly, and monitor for excessive daytime psychomotor impairment with any combined use. Use of other sedative-hypnotics during the night is not recommended. *Risk D: Consider Therapy Modification*

Fenfluramine: CNS Depressants may increase CNS depressant effects of Fenfluramine. *Risk C: Monitor*

Fexinidazole: May decrease serum concentrations of CYP2B6 Substrates (High risk with Inducers). Management: Avoid concomitant use of fexinidazole and CYP2B6 substrates when possible. If combined, monitor for reduced efficacy of the CYP2B6 substrate. *Risk D: Consider Therapy Modification*

Flibanserin: CNS Depressants may increase CNS depressant effects of Flibanserin. *Risk C: Monitor*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Flupentixol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Fluphenazine: CNS Depressants may increase CNS depressant effects of Fluphenazine. *Risk C: Monitor*

Gabapentin (Immediate Release): CNS Depressants may increase CNS depressant effects of Gabapentin (Immediate Release). *Risk C: Monitor*

Gabapentin Enacarbil (Extended Release): CNS Depressants may increase CNS depressant effects of Gabapentin Enacarbil (Extended Release). *Risk C: Monitor*

Ganaxolone: CNS Depressants may increase CNS depressant effects of Ganaxolone. *Risk C: Monitor*

Grapefruit Juice: May increase serum concentrations of Ketamine. *Risk C: Monitor*

GuanFACINE: CNS Depressants may increase CNS depressant effects of GuanFACINE. *Risk C: Monitor*

Haloperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

HydroXYZine: May increase CNS depressant effects of CNS Depressants. Management: Reduce the dose of hydroxyzine or other CNS depressants if these agents are combined due to the potential for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Iloperidone: CNS Depressants may increase CNS depressant effects of Iloperidone. *Risk C: Monitor*

Imipramine: CNS Depressants may increase CNS depressant effects of Imipramine. *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Isocarboxazid: CNS Depressants may increase adverse/toxic effects of Isocarboxazid. *Risk X: Avoid*

Ixabepilone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Lacosamide: CNS Depressants may increase CNS depressant effects of Lacosamide. *Risk C: Monitor*

Lasmiditan: CNS Depressants may increase CNS depressant effects of Lasmiditan. *Risk C: Monitor*

Lemborexant: May increase CNS depressant effects of CNS Depressants. Management: Dosage adjustments of lemborexant and of concomitant CNS depressants may be necessary when administered together because of potentially additive CNS depressant effects. Close monitoring for CNS depressant effects is necessary. *Risk D: Consider Therapy Modification*

LevETIRacetam: CNS Depressants may increase CNS depressant effects of LevETIRacetam. *Risk C: Monitor*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofepamine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Loxapine: CNS Depressants may increase CNS depressant effects of Loxapine. Management: Consider reducing the dose of CNS depressants administered concomitantly with loxapine due to an increased risk of respiratory depression, sedation, hypotension, and syncope. *Risk D: Consider Therapy Modification*

Lumacaftor and Ivacaftor: May decrease serum concentrations of CYP2B6 Substrates (High risk with Inducers). *Risk C: Monitor*

Lumateperone: CNS Depressants may increase CNS depressant effects of Lumateperone. *Risk C: Monitor*

Lurasidone: CNS Depressants may increase CNS depressant effects of Lurasidone. *Risk C: Monitor*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Meclizine: CNS depressant effects of CNS Depressants and Meclizine may increase with coadministration. *Risk C: Monitor*

Melitracen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Memantine: NMDA Receptor Antagonists may increase adverse/toxic effects of Memantine. *Risk C: Monitor*

Meprobamate: CNS depressant effects of CNS Depressants and Meprobamate may increase with coadministration. *Risk C: Monitor*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metaxalone: CNS depressant effects of CNS Depressants and Metaxalone may increase with coadministration. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methocarbamol: CNS Depressants may increase CNS depressant effects of Methocarbamol. *Risk C: Monitor*

Methotrimeprazine: CNS Depressants may increase CNS depressant effects of Methotrimeprazine. Methotrimeprazine may increase CNS depressant effects of CNS Depressants. Management: Reduce the usual dose of CNS depressants by 50% if starting methotrimeprazine until the dose of methotrimeprazine is stable. Monitor patient closely for evidence of CNS depression. *Risk D: Consider Therapy Modification*

Methoxyflurane: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methsuximide: CNS Depressants may increase CNS depressant effects of Methsuximide. *Risk C: Monitor*

Metoclopramide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

Mianserin: CNS Depressants may increase CNS depressant effects of Mianserin. *Risk C: Monitor*

MiFEPRIStone: May increase serum concentrations of CYP2B6 Substrates (High risk with Inhibitors). *Risk C: Monitor*

Milsaperidone: CNS Depressants may increase CNS depressant effects of Milsaperidone. *Risk C: Monitor*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mirogabalin: CNS Depressants may increase CNS depressant effects of Mirogabalin. *Risk C: Monitor*

Mirtazapine: CNS Depressants may increase CNS depressant effects of Mirtazapine. *Risk C: Monitor*

Mivacurium: Ketamine may increase therapeutic effects of Mivacurium. *Risk C: Monitor*

Molindone: CNS Depressants may increase CNS depressant effects of Molindone. *Risk C: Monitor*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Nalfurafine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nefopam: CNS Depressants may increase CNS depressant effects of Nefopam. *Risk C: Monitor*

Neuromuscular-Blocking Agents: Ketamine may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Nitrous Oxide: CNS Depressants may increase CNS depressant effects of Nitrous Oxide. *Risk C: Monitor*

Nortriptyline: CNS Depressants may increase CNS depressant effects of Nortriptyline. *Risk C: Monitor*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

OLANzapine: CNS depressant effects of OLANzapine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opipramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Orphenadrine: CNS Depressants may increase CNS depressant effects of Orphenadrine. *Risk C: Monitor*

Oxatomide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Oxybate (Mixed Salts and Sodium Salt): CNS Depressants may increase CNS depressant effects of Oxybate (Mixed Salts and Sodium Salt). Management: Consider alternatives to this combination when possible. If combined, dose reduction or discontinuation of one or more CNS depressants (including the oxybate salt product) should be considered. Interrupt oxybate salt treatment during short-term opioid use *Risk D: Consider Therapy Modification*

Paliperidone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Perampanel: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Perphenazine: CNS depressant effects of CNS Depressants and Perphenazine may increase with coadministration. *Risk C: Monitor*

Pheniramine: CNS Depressants may increase CNS depressant effects of Pheniramine. *Risk C: Monitor*

Phenyltoloxamine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pholcodine: CNS Depressants may increase CNS depressant effects of Pholcodine. *Risk C: Monitor*

Pimozide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pipamperone: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Pizotifen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pomalidomide: CNS Depressants may increase CNS depressant effects of Pomalidomide. *Risk C: Monitor*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Pregabalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Procarbazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Prochlorperazine: CNS Depressants may increase CNS depressant effects of Prochlorperazine. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Promethazine: CNS Depressants may increase CNS depressant effects of Promethazine. Management: Discontinue or reduce the dose of other CNS depressants during promethazine therapy. *Risk D: Consider Therapy Modification*

Propofol: CNS Depressants may increase CNS depressant effects of Propofol. *Risk C: Monitor*

Protriptyline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pyrilamine (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

QUetiapine: CNS Depressants may increase CNS depressant effects of QUetiapine. *Risk C: Monitor*

Ramelteon: CNS Depressants may increase CNS depressant effects of Ramelteon. *Risk C: Monitor*

Reserpine: CNS Depressants may increase CNS depressant effects of Reserpine. *Risk C: Monitor*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

RisperiDONE: CNS depressant effects of CNS Depressants and RisperiDONE may increase with coadministration. *Risk C: Monitor*

Ropeginterferon Alfa-2b: CNS Depressants may increase adverse/toxic effects of Ropeginterferon Alfa-2b. Specifically, the risk of neuropsychiatric adverse effects may be increased. Management: Avoid coadministration of ropeginterferon alfa-2b and other CNS depressants. If this combination cannot be avoided, monitor patients for neuropsychiatric adverse effects (eg, depression, suicidal ideation, aggression, mania). *Risk D: Consider Therapy Modification*

ROPINIrole: CNS Depressants may increase sedative effects of ROPINIrole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Scopolamine: CNS Depressants may increase CNS depressant effects of Scopolamine. Management: Avoid coadministration of scopolamine with other CNS depressants if possible. If coadministration is required, monitor patients for excessive CNS depression. *Risk D: Consider Therapy Modification*

Stiripentol: CNS Depressants may increase CNS depressant effects of Stiripentol. *Risk C: Monitor*

Suvorexant: CNS Depressants may increase CNS depressant effects of Suvorexant. *Risk C: Monitor*

Tasimelteon: CNS Depressants may increase CNS depressant effects of Tasimelteon. *Risk C: Monitor*

Tetrabenazine: CNS Depressants may increase CNS depressant effects of Tetrabenazine. *Risk C: Monitor*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

Theophylline Derivatives: Ketamine may increase adverse/toxic effects of Theophylline Derivatives. Specifically, the risk for seizures may be increased. *Risk C: Monitor*

Thioridazine: CNS Depressants may increase CNS depressant effects of Thioridazine. *Risk C: Monitor*

Thiotepa: May increase serum concentrations of CYP2B6 Substrates (High risk with Inhibitors). *Risk C: Monitor*

Thiothixene: CNS Depressants may increase CNS depressant effects of Thiothixene. *Risk C: Monitor*

Thyroid Products: May increase hypertensive effects of Ketamine. Thyroid Products may increase tachycardic effects of Ketamine. *Risk C: Monitor*

TiaGABine: CNS Depressants may increase CNS depressant effects of TiaGABine. *Risk C: Monitor*

Ticlopidine: May increase serum concentrations of Ketamine. *Risk C: Monitor*

TiZANidine: CNS Depressants may increase CNS depressant effects of TiZANidine. *Risk C: Monitor*

Tolcapone: CNS Depressants may increase CNS depressant effects of Tolcapone. *Risk C: Monitor*

Topiramate: CNS Depressants may increase CNS depressant effects of Topiramate. CNS Depressants may increase adverse/toxic effects of Topiramate. Specifically, cognitive dysfunction and neuropsychiatric adverse effects may be increased. *Risk C: Monitor*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

TraZODone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trifluoperazine: CNS Depressants may increase CNS depressant effects of Trifluoperazine. *Risk C: Monitor*

Trimeprazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trimethobenzamide: CNS Depressants may increase CNS depressant effects of Trimethobenzamide. Management: Avoid coadministration of trimethobenzamide with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients for CNS adverse effects (eg, depression, disorientation, seizures). *Risk D: Consider Therapy Modification*

Trimipramine: CNS Depressants may increase CNS depressant effects of Trimipramine. *Risk C: Monitor*

Triprolidine: CNS Depressants may increase CNS depressant effects of Triprolidine. *Risk C: Monitor*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Valproic Acid and Derivatives: CNS Depressants may increase CNS depressant effects of Valproic Acid and Derivatives. *Risk C: Monitor*

Vigabatrin: CNS Depressants may increase CNS depressant effects of Vigabatrin. *Risk C: Monitor*

Zaleplon: CNS Depressants may increase CNS depressant effects of Zaleplon. Management: Coadministration of zaleplon with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. If zaleplon is coadministered with other CNS depressants at other times, monitor for CNS depression; dose reductions may be required. *Risk D: Consider Therapy Modification*

Ziconotide: CNS Depressants may increase CNS depressant effects of Ziconotide. *Risk C: Monitor*

Ziprasidone: CNS Depressants may increase CNS depressant effects of Ziprasidone. *Risk C: Monitor*

Zolpidem: CNS Depressants may increase CNS depressant effects of Zolpidem. Management: Reduce the Intermezzo brand sublingual zolpidem adult dose to 1.75 mg for men who are also receiving other CNS depressants. No such dose change is recommended for women. Avoid use with other CNS depressants at bedtime; avoid use with alcohol. *Risk D: Consider Therapy Modification*

Zonisamide: CNS Depressants may increase CNS depressant effects of Zonisamide. *Risk C: Monitor*

Zopiclone: CNS Depressants may increase CNS depressant effects of Zopiclone. *Risk C: Monitor*

Zuclopenthixol: May increase CNS depressant effects of CNS Depressants. Management: Avoid use of high-dose hypnotics or other CNS depressants together with zuclopenthixol. *Risk D: Consider Therapy Modification*

Zuranolone: May increase CNS depressant effects of CNS Depressants. Management: Consider alternatives to the use of zuranolone with other CNS depressants or alcohol. If combined, consider a zuranolone dose reduction and monitor patients closely for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Reproductive Considerations

When treating depression, evaluate pregnancy status prior to initiating treatment in patients who could become pregnant. Treatment should not be withheld, but pharmacologic management may vary based on reproductive status, severity of illness, and history of antidepressant response (ACOG 2023; WFSBP [Dodd 2018]). Patients effectively treated may continue their current medication when planning a pregnancy unless contraindications exist (BAP [McAllister-Williams 2017]). Management of mental health conditions in patients who could become pregnant should be based on a shared decision-making process that considers the possibility of pregnancy during treatment (ACOG 2023; BAP [McAllister-Williams 2017]; CANMAT [MacQueen 2016]).

Pregnancy Considerations

Ketamine crosses the placenta (Ellingson 1977; Little 1972; Su 2010).

Ketamine produces dose-dependent increases in uterine contractions; effects may vary by trimester, with effects more common in early pregnancy. Dose-related neonatal depression and decreased Apgar scores have been reported with large doses administered at delivery; serum bilirubin may also be increased (Little 1972; Neuman 2013; White 1982). General hypotonia, resolving within 21 days, was observed in a newborn following maternal abuse of ketamine during pregnancy (Su 2010).

Based on animal data, repeated or prolonged use of general anesthetic and sedation medications that block N-methyl-D-aspartate (NMDA) receptors and/or potentiate gamma-aminobutyric acid (GABA) activity may affect brain development. Evaluate benefits and potential risks of fetal exposure to ketamine when duration of surgery is expected to be >3 hours (Olutoye 2018).

Due to pregnancy-induced physiologic changes, some pharmacokinetic properties of ketamine may be altered; clearance may be decreased (Little 1972).

Ketamine has been evaluated for anesthesia during cesarean and vaginal delivery and as an adjunctive analgesic in cesarean deliveries (ACOG 2019b; Corssen 1974; Heesen 2015; Wang 2020). Available data from ketamine use during cesarean delivery, although limited by retrospective nature, sample size, and lack of long-term follow-up, have not identified drug-associated risks of major birth defects, pregnancy loss, or other adverse maternal or fetal outcomes (manufacturer's labeling). When sedation and analgesia are needed for other procedures during pregnancy, low doses of ketamine may be used, but other agents are preferred (Neuman 2013; Schwenk 2018). The American College of Obstetricians and Gynecologists recommends that pregnant patients should not be denied medically necessary surgery, regardless of trimester. If the procedure is elective, it should be delayed until after delivery (ACOG 2019a).

Ketamine has been studied for the adjunctive treatment of depression (VA/DoD 2022). Untreated and undertreated mental health conditions are associated with adverse pregnancy outcomes including preterm birth, low birth weight, preeclampsia, postpartum depression, and impaired infant attachment (associated with long-term developmental effects). Discontinuing effective medications during pregnancy increases the risk of relapse. Management should be made as part of a shared decision-making process (ACOG 2023). Patients effectively treated for depression prepregnancy may use the same medication during pregnancy unless contraindications exist (ACOG 2023; CANMAT [MacQueen 2016]; BAP [McAllister-Williams 2017]). Treatment should not be withheld or discontinued based only on pregnancy status. When medications are used, the lowest effective dose of a single agent is recommended. Optimize dosing prior to changing a medication or adding additional agents whenever possible. Monthly monitoring for symptom improvement with a validated screening tool during pregnancy is recommended. Manage side effects as needed (ACOG 2023).

Ketamine infusion for the treatment of refractory status epilepticus in a pregnant patient has been noted in a case report (Talahma 2018).

Breastfeeding Considerations

Ketamine and its metabolites are present in human milk.

Multiple case reports provide data related to the presence of ketamine in breast milk:

- Ketamine and the active metabolite norketamine were observed in the breast milk of 4 lactating patients evaluated between 1 and 23 months postpartum. All patients in this case series were administered intermittent ketamine infusions for the treatment of depression. Ketamine doses ranged from 0.607 to 4.167 mg/kg (49 to 378 mg). Patients provided breast milk samples prior to and at specified intervals for 24 hours following a dose. Maximum breast milk concentrations of ketamine and norketamine occurred 1 hour after the start of the infusion. Authors of the study calculated the estimated infant dose of ketamine via breast milk to be 0.003 mg/kg/day to 0.017 mg/kg/day providing a relative infant dose (RID) of 0.34% to 0.57% compared to the weight-adjusted maternal dose. The estimated infant dose of norketamine via breast milk was calculated to be 0.005 mg/kg/day to 0.018 mg/kg/day providing an RID of 0.29% to 0.95%. Participants did not observe adverse events in the breastfed infants during the study period (Majdinasab 2023).
- Ketamine, norketamine, and dehydronorketamine were detected in breast milk following an infusion of ketamine 0.1 mg/kg/hour over 12 hours (maximum 8 mg/hour) to lactating patients ~6 months

postpartum. Breast milk was sampled at 7 specified intervals between 2 and 20 hours after the start of the infusion. Ketamine concentrations in breast milk were greater than the metabolites at all time points. Using data from 1 patient, the highest ketamine concentration occurred at ~8 hours into the infusion. It was not noted if infants were breastfed (Shahane 2023).

- Data are available from 4 lactating patients following single injections of ketamine 0.5 mg/kg IM and 1 mg/kg SUBQ. Patients were at least 3 months postpartum, doses were separated by 5 to 14 days, and breast milk was collected prior to and at 3-hour intervals after the dose. Ketamine, norketamine, dehydronorketamine, and hydroxynorketamine were detected in breast milk. Authors of the study calculated the mean estimated infant dose of ketamine and metabolites via breast milk to be 0.00766 mg/kg over 12 hours providing a RID of 0.766% compared to the 1 mg/kg weight-adjusted maternal dose. The RID calculated using the 0.5 mg/kg maternal dose was 0.65%. Infants did not receive breast milk in this study (Wolfson 2023).

- In general, breastfeeding is considered acceptable when the RID of a medication is <10% (Anderson 2016; Ito 2000); however, some sources note breastfeeding should only be considered if the RID is <5% for psychotropic agents (Anderson 2021).

Adverse events, including clinically significant weight loss or jaundice, were not observed in infants of 298 mothers who received ketamine as an anesthetic for tubal ligation. Maternal doses of ketamine were ≤ 3.8 mg/kg, patients were between 1 and 6 days postpartum, and most infants were fully breastfed (Gilder 2021). According to the manufacturer, limit ketamine treatment in breastfeeding patients to anesthesia due to the potential for developmental neurotoxicity; monitor infants for sedation, respiratory depression, and increased muscle tone and spasms. Consider the health benefits, the patient's clinical needs, and the potential for adverse effects prior to administering ketamine.

The Academy of Breastfeeding Medicine recommends postponing elective surgery until milk supply and breastfeeding are established. Milk should be expressed ahead of surgery when possible. In general, when the child is healthy and full term, breastfeeding may resume, or milk may be expressed once the mother is awake and in recovery. For children who are at risk for apnea, hypotension, or hypotonia, milk may be saved for later use when the child is at lower risk (ABM [Reece-Stremtan 2017]).

Small supplemental doses of ketamine used during cesarean delivery should not prevent breastfeeding once the mother is stable and alert; however, data are insufficient to recommend use for long-term pain control (ABM [Martin 2018]).

Patients effectively treated for depression during pregnancy may continue their medication postpartum unless contraindications to breastfeeding exist. The presence and concentration of the drug in breast milk, efficacy of maternal treatment, and infant age should be considered when initiating a medication for the first time postpartum (ABM [Sriraman 2015]; BAP [McAllister-Williams 2017]).

Monitoring Parameters

Heart rate, blood pressure, respiratory rate, transcutaneous O₂ saturation, level of sedation, emergence reactions; cardiac function (continuously monitored in patients with increased blood pressure or cardiac decompensation); LFTs, alkaline phosphatase, and gamma glutamyl transferase (baseline and then at periodic intervals); signs or symptoms of urinary obstruction or severe lower urinary tract symptoms. When used for the treatment of pain, monitor pain control during therapy.

Mechanism of Action

Produces a cataleptic-like state in which the patient is dissociated from the surrounding environment by direct action on the cortex and limbic system. Ketamine is a noncompetitive NMDA receptor antagonist that blocks glutamate. Low (subanesthetic) doses produce analgesia, and modulate central sensitization, hyperalgesia and opioid tolerance. Reduces polysynaptic spinal reflexes.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action:

IV: Anesthetic effect: Within 30 seconds.

IM: Anesthetic effect: 3 to 4 minutes; Analgesia: Within 10 to 15 minutes.

Intranasal: Analgesic effect: Within 10 minutes (Carr 2004); Sedation: Children 2 to 6 years: 5 to 8 minutes (Bahetwar 2011).

Oral: Analgesia: Within 30 minutes; Sedation: Children 2 to 8 years (Turhanoglu 2003):

4 mg/kg/dose: 12.9 ± 1.9 minutes.

6 mg/kg/dose: 10.4 ± 2.9 minutes.

8 mg/kg/dose: 9.5 ± 1.9 minutes.

Duration:

IV: Anesthetic effect: 5 to 10 minutes; Recovery: 1 to 2 hours.

IM: Anesthetic effect: 12 to 25 minutes; Analgesia: 15 to 30 minutes; Recovery: 3 to 4 hours.

Intranasal: Analgesic effect: Up to 60 minutes (Carr 2004); Recovery: Children 2 to 6 years: 34 to 46 minutes (Bahetwar 2011).

Distribution: V_{dss} : 2.1 to 3.1 L/kg (Clements 1981).

Protein binding: 10% to 50% (Zanos 2018).

Metabolism: Extensive hepatic metabolism via N-demethylation during P-450 metabolism (primarily via CYP2B6 and CYP3A4) to form the primary metabolite, norketamine (80%), a pharmacologically active metabolite with approximately one-third of the analgesic potency of ketamine. Norketamine is further metabolized primarily to hydroxynorketamine (HNK) and other hydroxylated metabolites, and to a lesser extent, dehydronorketamine (DHNK). Both HNK and DHNK may possess some degree of pharmacologic activity, but the clinical implications have not been fully defined. Other minor metabolic pathways of ketamine exist but are less studied (Mion 2013; Zanos 2018). When administered orally, norketamine concentrations are higher compared to other routes of administration due to extensive first-pass metabolism in the liver (Blonk 2010; Soto 2012).

Bioavailability:

IM: 93%.

Oral: 20% to 30% (Miller 2010).

Intranasal: Children, Adolescents, and Adults: Mean range: 35% to 50% (Malinovsky 1996; Miller 2010; Nielsen 2014).

Rectal: Children 2 to 9 years: 25% (Malinovsky 1996).

Half-life elimination: Alpha: 10 to 15 minutes; Beta: 2.5 hours.

Time to peak, plasma:

IM: 5 to 30 minutes (Clements 1982).

Intranasal: 10 to 14 minutes (Huge 2010); Children 2 to 9 years: ~20 minutes (Malinovsky 1996).

Oral: ~30 minutes (Soto 2012).

Rectal: Children 2 to 9 years: ~45 minutes (Malinovsky 1996).

Excretion: Urine (91%); feces (3%) (Ghoneim 1977).

Patient Counseling Points

(For additional information see "Ketamine: Patient drug information")

What is this drug used for?

- It is used for anesthesia before or during some procedures or surgery.

- This drug may be used for other reasons. If you have been given this drug for some other reason, talk with your doctor about the benefits and risks. Talk with your doctor if you have questions or concerns about taking this drug.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Feeling sleepy
- Decreased appetite
- Upset stomach or throwing up
- Irritation where this drug is given

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- High or low blood pressure like very bad headache or dizziness, passing out, or change in eyesight
- Liver or bile duct problems like dark urine, tiredness, decreased appetite, upset stomach or stomach pain, light-colored stools, throwing up, or yellow skin or eyes
- Trouble breathing, slow breathing, or shallow breathing
- Fast, slow, or abnormal heartbeat
- Shortness of breath, a big weight gain, or swelling in the arms or legs
- Feeling confused
- Hallucinations (seeing or hearing things that are not there)
- Change in how you act
- Mood changes
- Muscle stiffness
- Seizures
- Change in eyesight
- Not able to control eye movements
- Trouble passing urine, pain when passing urine, passing urine in a weak stream or drips, or passing urine more often
- Blood in the urine
- Pelvis or side pain
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to

provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Ketalar;
- (AR) Argentina: Inducmina;
- (AT) Austria: Ketalar;
- (AU) Australia: Ketamine interpharma;
- (BD) Bangladesh: Kain;
- (BE) Belgium: Ketalar;
- (CH) Switzerland: Ketamin sintetica;
- (CL) Chile: Ketalar;
- (CZ) Czech Republic: Calypsol | Ketalar | Narkamon;
- (DE) Germany: Ketamin curamed;
- (EC) Ecuador: Ketalar;
- (HU) Hungary: Ketanest;
- (IL) Israel: Ketalar;
- (JP) Japan: Ketalar;
- (LT) Lithuania: Ketalar | Narkamon;
- (LV) Latvia: Ketalar | Narkamon;
- (NZ) New Zealand: Ketamine;
- (PE) Peru: Ketalar;
- (PK) Pakistan: Calypsol | Ketlar;
- (PL) Poland: Ketalar | Ketamin Inresa | Narkamon;
- (PR) Puerto Rico: Ketamine HCL;
- (QA) Qatar: Ketalar | Tekam;
- (RU) Russian Federation: Ketalar | Ketamin;
- (SI) Slovenia: Ketanest;
- (SK) Slovakia: Calypsol | Narkamon;
- (TW) Taiwan: Ketalar | Ketomin;
- (UA) Ukraine: Ketalar;
- (VE) Venezuela, Bolivarian Republic of: Keiran;
- (ZA) South Africa: Micro ketamine

Use of UpToDate is subject to the Terms of Use.

REFERENCES

1. aan het Rot M, Collins KA, Murrough JW, et al. Safety and efficacy of repeated-dose intravenous ketamine for treatment-resistant depression. *Biol Psychiatry*. 2010;67(2):139-145. doi:10.1016/j.biopsych.2009.08.038 [PubMed 19897179]
2. Acquisto NM, Mosier JM, Bittner EA, et al. Society of Critical Care Medicine clinical practice guidelines for rapid sequence intubation in the critically ill adult patient. *Crit Care Med*. 2023;51(10):1411-1430. doi:10.1097/CCM.0000000000006000 [PubMed 37707379]
3. Adhikary SD, Thiruvengkatarajan V, McFadden A, Liu WM, Mets B, Rogers A. Analgesic efficacy of ketamine and magnesium after laparoscopic sleeve gastrectomy: a randomized, double-blind, placebo-controlled trial. *J Clin Anesth*. 2021;68:110097. doi:10.1016/j.jclinane.2020.110097 [PubMed 33120301]

4. Ahuja S, Brendle M, Smart L, Moore C, Thielking P, Robison R. Real-world depression, anxiety and safety outcomes of intramuscular ketamine treatment: a retrospective descriptive cohort study. *BMC Psychiatry*. 2022;22(1):634. doi:10.1186/s12888-022-04268-5 [PubMed 36192794]
5. Akhlaghi N, Payandemehr P, Yaseri M, Akhlaghi AA, Abdolrazaghnejad A. Premedication with midazolam or haloperidol to prevent recovery agitation in adults undergoing procedural sedation with ketamine: a randomized double-blind clinical trial. *Ann Emerg Med*. 2019;73(5):462-469. doi:10.1016/j.annemergmed.2018.11.016 [PubMed 30611640]
6. Albanèse J, Arnaud S, Rey M, Thomachot L, Alliez B, Martin C. Ketamine decreases intracranial pressure and electroencephalographic activity in traumatic brain injury patients during propofol sedation. *Anesthesiology*. 1997;87(6):1328-1334. doi:10.1097/00000542-199712000-00011 [PubMed 9416717]
7. Alletag MJ, Auerbach MA, Baum CR. Ketamine, propofol, and ketofol use for pediatric sedation. *Pediatr Emerg Care*. 2012;28(12):1391-1395; quiz 1396-8. doi:10.1097/PEC.0b013e318276fde2. [PubMed 23222112]
8. American College of Obstetricians and Gynecologists (ACOG). ACOG committee opinion no. 775: nonobstetric surgery during pregnancy. *Obstet Gynecol*. 2019a;133(4):e285-e286. doi:10.1097/AOG.0000000000003174 [PubMed 30913200]
9. American College of Obstetricians and Gynecologists (ACOG). ACOG practice bulletin no. 209: obstetric analgesia and anesthesia. *Obstet Gynecol*. 2019b;133(3):e208-e225. doi:10.1097/AOG.0000000000003132 [PubMed 30801474]
10. American College of Obstetricians and Gynecologists. Treatment and management of mental health conditions during pregnancy and postpartum: ACOG clinical practice guideline no. 5. *Obstet Gynecol*. 2023;141(6):1262-1288. doi:10.1097/AOG.0000000000005202 [PubMed 37486661]
11. American Society of Health-System Pharmacists (ASHP). Pediatric continuous infusion standards. <https://www.ashp.org/-/media/assets/pharmacy-practice/s4s/docs/Pediatric-Infusion-Standards.ashx>. Updated March 2024. Accessed May 28, 2024.
12. Anand KJ, International Evidence-Based Group for Neonatal Pain. Consensus statement for the prevention and management of pain in the newborn. *Arch Pediatr Adolesc Med*. 2001;155(2):173-180. doi:10.1001/archpedi.155.2.173 [PubMed 11177093]
13. Anderson PO. Antidepressants and breastfeeding. *Breastfeed Med*. 2021;16(1):5-7. doi:10.1089/bfm.2020.0350 [PubMed 33237799]
14. Anderson PO, Sauberan JB. Modeling drug passage into human milk. *Clin Pharmacol Ther*. 2016;100(1):42-52. doi:10.1002/cpt.377 [PubMed 27060684]
15. Andolfatto G, Abu-Laban RB, Zed PJ, et al. Ketamine-propofol combination (ketofol) versus propofol alone for emergency department procedural sedation and analgesia: a randomized double-blind trial. *Ann Emerg Med*. 2012;59(6):504-512.e1-e2. doi:10.1016/j.annemergmed.2012.01.017 [PubMed 22401952]
16. Andolfatto G, Willman E, Joo D, et al. Intranasal ketamine for analgesia in the emergency department: a prospective observational series. *Acad Emerg Med*. 2013;20(10):1050-1054. doi:10.1111/acem.12229 [PubMed 24127709]
17. Andrade C. Ketamine for depression, 4: in what dose, at what rate, by what route, for how long, and at what frequency? *J Clin Psychiatry*. 2017;78(7):e852-e857. doi:10.4088/JCP.17f11738 [PubMed 28749092]
18. Aroni F, Iacovidou N, Dontas I, Pourzitaki C, Xanthos T. Pharmacological aspects and potential new clinical applications of ketamine: reevaluation of an old drug. *J Clin Pharmacol*. 2009;49(8):957-964. doi:10.1177/0091270009337941 [PubMed 19546251]

19. Asadi P, Ghafouri HB, Yasinzadeh M, Kasnavieh SM, Modirian E. Ketamine and atropine for pediatric sedation: a prospective double-blind randomized controlled trial. *Pediatr Emerg Care*. 2013;29(2):136-139. doi:10.1097/PEC.0b013e31828058b2 [PubMed 23364373]
20. Bahetwar SK, Pandey RK, Saksena AK, Chandra G. A comparative evaluation of intranasal midazolam, ketamine and their combination for sedation of young uncooperative pediatric dental patients: a triple blind randomized crossover trial. *J Clin Pediatr Dent*. 2011;35(4):415-420. doi:10.17796/jcpd.35.4.l43h3354705u2574 [PubMed 22046702]
21. Ballow SL, Kaups KL, Anderson S, Chang M. A standardized rapid sequence intubation protocol facilitates airway management in critically injured patients. *J Trauma Acute Care Surg*. 2012;73(6):1401-1405. doi:10.1097/TA.0b013e318270dcf5 [PubMed 23188232]
22. Barkan S, Breitbart R, Brenner-Zada G, et al. A double-blind, randomised, placebo-controlled trial of oral midazolam plus oral ketamine for sedation of children during laceration repair. *Emerg Med J*. 2014;31(8):649-653. doi:10.1136/emered-2012-202189 [PubMed 23686730]
23. Barnard EB, Moy RJ, Kehoe AD, Bebartá VS, Smith JE. Rapid sequence induction of anaesthesia via the intraosseous route: a prospective observational study. *Emerg Med J*. 2015;32(6):449-452. doi:10.1136/emered-2014-203740 [PubMed 24963149]
24. Barois J, Tourneux P. Ketamine and atropine decrease pain for preterm newborn tracheal intubation in the delivery room: an observational pilot study. *Acta Paediatr*. 2013;102(12):e534-538. doi:10.1111/apa.12413 [PubMed 24015945]
25. Barr J, Fraser GL, Puntillo K, et al. Clinical practice guidelines for the management of pain, agitation, and delirium in adult patients in the intensive care unit. *Crit Care Med*. 2013;41(1):263-306. doi:10.1097/CCM.0b013e3182783b72 [PubMed 23269131]
26. Based on expert opinion.
27. Beaudequin D, Can AT, Dutton M, et al. Predicting therapeutic response to oral ketamine for chronic suicidal ideation: a Bayesian network for clinical decision support. *BMC Psychiatry*. 2020;20(1):519. doi:10.1186/s12888-020-02925-1 [PubMed 33115424]
28. Beaudrie-Nunn AN, Wieruszewski ED, Woods EJ, Bellolio F, Mara KC, Canterbury EA. Efficacy of analgesic and sub-dissociative dose ketamine for acute pain in the emergency department. *Am J Emerg Med*. 2023;70:133-139. doi:10.1016/j.ajem.2023.05.026 [PubMed 37290249]
29. Beebe DS, Belani KG, Chang PN, et al. Effectiveness of preoperative sedation with rectal midazolam, ketamine, or their combination in young children. *Anesth Analg*. 1992;75(6):880-884. doi:10.1213/00000539-199212000-00003 [PubMed 1443705]
30. Berkenbosch JW, Graff GR, Stark JM. Safety and efficacy of ketamine sedation for infant flexible fiberoptic bronchoscopy. *Chest*. 2004;125(3):1132-1137. doi:10.1378/chest.125.3.1132 [PubMed 15006978]
31. Bilbao-Meseguer I, Rodríguez-Gascón A, Barrasa H, Isla A, Solinís MÁ. Augmented renal clearance in critically ill patients: a systematic review. *Clin Pharmacokinet*. 2018;57(9):1107-1121. doi:10.1007/s40262-018-0636-7 [PubMed 29441476]
32. Blonk MI, Koder BG, van den Bemt PM, Huygen FJ. Use of oral ketamine in chronic pain management: a review. *Eur J Pain*. 2010;14(5):466-472. doi:10.1016/j.ejpain.2009.09.005 [PubMed 19879174]
33. Bovill JG. Intravenous anesthesia for the patient with left ventricular dysfunction. *Semin Cardiothorac Vasc Anesth*. 2006;10(1):43-48. doi: 10.1177/108925320601000108 [PubMed 16703233]
34. Bowles ED, Gold ME. Rethinking the paradigm: evaluation of ketamine as a neurosurgical anesthetic [published correction appears in *AANA J*. 2013;81(1):18.]. *AANA J*. 2012;80(6):445-452. [PubMed 23409639]

35. Bravenec B. General anesthesia: Intravenous induction agents. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed July 29, 2020.
36. Brenner LA, Capaldi V, Constans J, et al. Assessment and management of patients at risk for suicide: synopsis of the 2024 U.S. Department of Veterans Affairs and U.S. Department of Defense clinical practice guidelines. *Ann Intern Med*. 2025;178(3):416-425. doi:10.7326/ANNALS-24-01938 [PubMed 39903866]
37. Brophy GM, Bell R, Claassen J, et al; Neurocritical Care Society Status Epilepticus Guideline Writing Committee. Guidelines for the evaluation and management of status epilepticus. *Neurocrit Care*. 2012;17(1):3-23. doi:10.1007/s12028-012-9695-z [PubMed 22528274]
38. Brunton LL, Lazo JS, Park KL, eds. *Goodman & Gilman's The Pharmacological Basis of Therapeutics*. 11th ed. McGraw-Hill Medical; 2006.
39. Buchheit JL, Yeh DD, Eikermann M, Lin H. Impact of low-dose ketamine on the usage of continuous opioid infusion for the treatment of pain in adult mechanically ventilated patients in surgical intensive care units. *J Intensive Care Med*. 2019;34(8):646-651. doi:10.1177/0885066617706907 [PubMed 28468568]
40. Cabbabe EB, Behbahani PM. Cardiovascular reactions associated with the use of ketamine and epinephrine in plastic surgery. *Ann Plast Surg*. 1985;15(1):50-56. doi:10.1097/00000637-198507000-00006 [PubMed 2417535]
41. Carr DB, Goudas LC, Denman WT, et al. Safety and efficacy of intranasal ketamine for the treatment of breakthrough pain in patients with chronic pain: a randomized, double-blind, placebo-controlled, crossover study. *Pain*. 2004;108(1-2):17-27. doi:10.1016/j.pain.2003.07.001 [PubMed 15109503]
42. Castilla M, Jerez M, Llácer M, Martinez S. Anaesthetic management in a neonate with congenital complete heart block. *Paediatr Anaesth*. 2004;14(2):172-175. doi:10.1046/j.1460-9592.2003.01180.x [PubMed 14962334]
43. Chen CH, Lee MH, Chen YC, Lin MF. Ketamine-snorting associated cystitis. *J Formos Med Assoc*. 2011;110(12):787-791. doi:10.1016/j.jfma.2011.11.010 [PubMed 22248834]
44. Cherng CH, Wong CS. Ketamine-induced emergence reactions after desflurane anaesthesia. *Eur J Anaesthesiol*. 2007;24(2):200-201. doi:10.1017/S0265021506001773 [PubMed 17038218]
45. Chilukuri H, Reddy NP, Pathapati RM, Manu AN, Jollu S, Shaik AB. Acute antidepressant effects of intramuscular versus intravenous ketamine. *Indian J Psychol Med*. 2014;36(1):71-76. doi:10.4103/0253-7176.127258 [PubMed 24701015]
46. Chow TG, McConnell J, Lee MJ. A 2-year-old girl with periprocedural anaphylaxis. *Allergy Asthma Proc*. 2021;42(1):97-99. doi:10.2500/aap.2021.42.200031 [PubMed 33404393]
47. Chudnofsky CR, Weber JE, Stoyanoff PJ, et al. A combination of midazolam and ketamine for procedural sedation and analgesia in adult emergency department patients. *Acad Emerg Med*. 2000;7(3):228-235. doi:10.1111/j.1553-2712.2000.tb01064.x [PubMed 10730829]
48. Clements JA, Nimmo WS, Grant IS. Bioavailability, pharmacokinetics, and analgesic activity of ketamine in humans. *J Pharm Sci*. 1982;71(5):539-542. doi:10.1002/jps.2600710516 [PubMed 7097501]
49. Clements JA, Nimmo WS. Pharmacokinetics and analgesic effect of ketamine in man. *Br J Anaesth*. 1981;53(1):27-30. doi:10.1093/bja/53.1.27 [PubMed 7459184]
50. Cohen L, Athaide V, Wickham ME, Doyle-Waters MM, Rose NG, Hohl CM. The effect of ketamine on intracranial and cerebral perfusion pressure and health outcomes: a systematic review. *Ann Emerg Med*. 2015;65(1):43-51.e2. doi:10.1016/j.annemergmed.2014.06.018 [PubMed 25064742]
51. Cohen SP, Bhatia A, Buvanendran A, et al. Consensus guidelines on the use of intravenous ketamine infusions for chronic pain from the American Society of Regional Anesthesia and Pain

- Medicine, the American Academy of Pain Medicine, and the American Society of Anesthesiologists. *Reg Anesth Pain Med*. 2018;43(5):521-546. doi:10.1097/AAP.0000000000000808 [PubMed 29870458]
52. Cole JB, Klein LR, Nystrom PC, et al. A prospective study of ketamine as primary therapy for pre-hospital profound agitation. *Am J Emerg Med*. 2018;36(5):789-796. doi:10.1016/j.ajem.2017.10.022 [PubMed 29033344]
 53. Cole JB, Moore JC, Nystrom PC, et al. A prospective study of ketamine versus haloperidol for severe prehospital agitation. *Clin Toxicol (Phila)*. 2016;54(7):556-562. doi:10.1080/15563650.2016.1177652 [PubMed 27102743]
 54. Corrigan M, Wilson SS, Hampton J. Safety and efficacy of intranasally administered medications in the emergency department and prehospital settings. *Am J Health Syst Pharm*. 2015;72(18):1544-1554. doi:10.2146/ajhp140630 [PubMed 26346210]
 55. Corssen G. Ketamine in obstetric anesthesia. *Clin Obstet Gynecol*. 1974;17(2):249-258. doi:10.1097/00003081-197406000-00015 [PubMed 4844201]
 56. Cosci F, Chouinard G. Acute and persistent withdrawal syndromes following discontinuation of psychotropic medications. *Psychother Psychosom*. 2020;89(5):283-306. doi:10.1159/000506868 [PubMed 32259826]
 57. Cote CJ. Sedation for the Pediatric Patient: A Review. *Pediatr Clin North Am*. 1994;41(1):31-58. doi:10.1016/s0031-3955(16)38690-4 [PubMed 8295806]
 58. Coté CJ, Lerman J, Anderson BJ, eds. *A Practice of Anesthesia for Infants and Children*. 5th ed. Elsevier Saunders; 2013.
 59. Coté CJ, Lerman J, Anderson B, eds. *A Practice of Anesthesia for Infants and Children*. 6th ed. Elsevier; 2019.
 60. Cotter S, Wong J, Gada N, et al. Repeated or continuous medically supervised ketamine administration associated with hepatobiliary adverse events: a retrospective case series. *Drug Saf*. 2021;44(12):1365-1374. doi:10.1007/s40264-021-01120-9 [PubMed 34699023]
 61. Critchlow DG. A case of ketamine dependence with discontinuation symptoms. *Addiction*. 2006;101(8):1212-1213. doi:10.1111/j.1360-0443.2006.01494.x [PubMed 16869851]
 62. Cunningham AJ, Barry P. Intraocular pressure--physiology and implications for anaesthetic management. *Can Anaesth Soc J*. 1986;33(2):195-208. doi:10.1007/BF03010831 [PubMed 3516335]
 63. David H, Shipp J. A randomized controlled trial of ketamine/propofol versus propofol alone for emergency department procedural sedation. *Ann Emerg Med*. 2011;57(5):435-441. doi:10.1016/j.annemergmed.2010.11.025 [PubMed 21256626]
 64. Davis J, Bates L. Rapid sequence induction via an intraosseous needle. *J Intensive Care Soc*. 2016;17(2):178-179. doi:10.1177/1751143715614845 [PubMed 28979485]
 65. Denmark TK, Crane HA, Brown L. Ketamine to avoid mechanical ventilation in severe pediatric asthma. *J Emerg Med*. 2006;30(2):163-166. doi:10.1016/j.jemermed.2005.09.003 [PubMed 16567251]
 66. Devlin JW, Skrobik Y, Gélinas C, et al; Society of Critical Care Medicine. Clinical practice guidelines for the prevention and management of pain, agitation/sedation, delirium, immobility, and sleep disruption in adult patients in the ICU. *Crit Care Med*. 2018;46(9):e825-e873. doi:10.1097/CCM.0000000000003299 [PubMed 30113379]
 67. Diaz JH. Intranasal ketamine preinduction of paediatric outpatients. *Paediatr Anaesth*. 1997;7(4):273-278. doi:10.1046/j.1460-9592.1997.d01-93.x [PubMed 9243683]
 68. Dodd S, Mitchell PB, Bauer M, et al. Monitoring for antidepressant-associated adverse events in the treatment of patients with major depressive disorder: an international consensus statement.

- World J Biol Psychiatry*. 2018;19(5):330-348. doi:10.1080/15622975.2017.1379609 [PubMed 28984491]
69. Drayna PC, Estrada C, Wang W, Saville BR, Arnold DH. Ketamine sedation is not associated with clinically meaningful elevation of intraocular pressure. *Am J Emerg Med*. 2012;30(7):1215-1218. doi:10.1016/j.ajem.2011.06.001. [PubMed 22169582]
 70. Driver BE, Reardon RF. Apnea after low-dose ketamine sedation during attempted delayed sequence intubation. *Ann Emerg Med*. 2017;69(1):34-35. doi:10.1016/j.annemergmed.2016.07.026 [PubMed 27692755]
 71. Ellingson A, Haram K, Sagen N, Solheim E. Transplacental passage of ketamine after intravenous administration. *Acta Anaesthesiol Scand*. 1977;21(1):41-44. doi:10.1111/j.1399-6576.1977.tb01191.x [PubMed 842268]
 72. El Sharkawy RA. Efficacy of adding low-dose ketamine to dexmedetomidine versus low-dose ketamine and propofol for conscious sedation in patients undergoing awake fiber-optic intubation. *Anesth Essays Res*. 2019;13(1):73-78. doi:10.4103/aer.AER_181_18 [PubMed 31031484]
 73. Erstad BL, Barletta JF. Drug dosing in the critically ill obese patient-a focus on sedation, analgesia, and delirium. *Crit Care*. 2020;24(1):315. doi:10.1186/s13054-020-03040-z [PubMed 32513237]
 74. Expert opinion. Senior Renal Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.
 75. Fantacci C, Fabrizio GC, Ferrara P, Franceschi F, Chiaretti A. Intranasal drug administration for procedural sedation in children admitted to pediatric emergency room. *Eur Rev Med Pharmacol Sci*. 2018;22(1):217-222. doi:10.26355/eurev_201801_14120 [PubMed 29364490]
 76. Fava M, Freeman MP, Flynn M, et al. Double-blind, placebo-controlled, dose-ranging trial of intravenous ketamine as adjunctive therapy in treatment-resistant depression (TRD). *Mol Psychiatry*. 2020;25(7):1592-1603. doi:10.1038/s41380-018-0256-5 [PubMed 30283029]
 77. FDA Drug Safety Communication. FDA review results in new warnings about using general anesthetics and sedation drugs in young children and pregnant women. Food and Drug Administration website. <https://www.fda.gov/Drugs/DrugSafety/ucm532356.htm>. Accessed May 26, 2017.
 78. Ferguson CL, Beckett RD. Intranasal ketamine for treatment of acute pain in pediatrics: a systematic review. *Pediatr Emerg Care*. 2020;36(8):e476-e481. doi:10.1097/PEC.0000000000002181 [PubMed 32665505]
 79. Fishman DS, Andropoulos DB, Lightdale JR. Sedation and the Food and Drug Administration warning: what a pediatric gastroenterologist, hepatologist, and pancreatologist should know. *J Pediatr Gastroenterol Nutr*. 2019;69(1):3-5. doi:10.1097/MPG.0000000000002346 [PubMed 30921252]
 80. Fond G, Loundou A, Rabu C, et al. Ketamine administration in depressive disorders: a systematic review and meta-analysis. *Psychopharmacology (Berl)*. 2014;231(18):3663-3676. doi:10.1007/s00213-014-3664-5 [PubMed 25038867]
 81. Frey TM, Florin TA, Caruso M, Zhang N, Zhang Y, Mittiga MR. Effect of intranasal ketamine vs fentanyl on pain reduction for extremity injuries in children: the PRIME randomized clinical trial. *JAMA Pediatr*. 2019;173(2):140-146. doi:10.1001/jamapediatrics.2018.4582 [PubMed 30592476]
 82. Fuchs B. Sedative-analgesia in ventilated adults: medication properties, dose regimens, and adverse effects. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed August 27, 2024.
 83. Fuhrman B, Zimmerman J, eds. *Pediatric Critical Care*. 4th ed. Elsevier Health; 2011.
 84. Garber PM, Droege CA, Carter KE, Harger NJ, Mueller EW. Continuous infusion ketamine for adjunctive analgo-sedation in mechanically ventilated, critically ill patients. *Pharmacotherapy*.

2019;39(3):288-296. doi:10.1002/phar.2223 [PubMed 30746728]

85. Garg A, Sinha P, Kumar P, Prakash O. Use of naltrexone in ketamine dependence. *Addict Behav.* 2014;39(8):1215-1216. doi:10.1016/j.addbeh.2014.04.004 [PubMed 24813545]
86. Gaspard N, Foreman B, Judd LM, et al. Intravenous ketamine for the treatment of refractory status epilepticus: a retrospective multicenter study. *Epilepsia.* 2013;54(8):1498-1503. doi:10.1111/epi.12247 [PubMed 23758557]
87. Gautam SN, Bhatta S, Sangraula D, Shrestha BC, Rawal SB. Intranasal midazolam vs ketamine as premedication in paediatric surgical procedure for child separation and induction. *Nepal Med Coll J.* 2007;9(3):179-181. [PubMed 18092436]
88. Geiger-Hayes J. PAINWeek journal vol 6, Q4. <https://www.painweek.org/media/journal/pwj-painweek-journal-vol-6-q4>. Published 2018. Accessed July 29, 2020.
89. Gelissen HP, Epema AH, Henning RH, Krijnen HJ, Hennis PJ, den Hertog A. Inotropic effects of propofol, thiopental, midazolam, etomidate, and ketamine on isolated human atrial muscle. *Anesthesiology.* 1996;84(2):397-403. doi:10.1097/00000542-199602000-00019 [PubMed 8602672]
90. Ghoneim MM, Korttila K. Pharmacokinetics of Intravenous Anaesthetics: Implications for Clinical Use. *Clin Pharmacokinet.* 1977;2(5):344-372. doi:10.2165/00003088-197702050-00003 [PubMed 21053]
91. Gilder ME, Tun NW, Carter A, et al. Outcomes for 298 breastfed neonates whose mothers received ketamine and diazepam for postpartum tubal ligation in a resource-limited setting. *BMC Pregnancy Childbirth.* 2021;21(1):121. doi:10.1186/s12884-021-03610-1 [PubMed 33563234]
92. Ginsburg AD, Heaton HA, Beaudrie-Nunn A, et al. High- versus low-dose ketamine for analgesia in older adults in the emergency department. *Am J Emerg Med.* 2024;86:120-124. doi:10.1016/j.ajem.2024.10.015 [PubMed 39427499]
93. Glue P, Neehoff S, Beaglehole B, Shadli S, McNaughton N, Hughes-Medlicott NJ. Ketamine for treatment-resistant major depressive disorder: double-blind active-controlled crossover study. *J Psychopharmacol.* 2024;38(2):162-167. doi:10.1177/02698811241227026 [PubMed 38293803]
94. Godwin SA, Burton JH, Gerardo CJ, et al; American College of Emergency Physicians. Clinical policy: procedural sedation and analgesia in the emergency department. *Ann Emerg Med.* 2014;63(2):247-258.e18. doi:10.1016/j.annemergmed.2013.10.015 [PubMed 24438649]
95. Gómez-Revuelta M, Fernández-Rodríguez M, Boada-Antón L, Vázquez-Bourgon J. Apnea during slow sub-anaesthetic infusion of intravenous ketamine for treatment-resistant depression. *Ther Adv Psychopharmacol.* 2020;10:2045125320981498. doi:10.1177/2045125320981498 [PubMed 33489089]
96. Goyal S, Ambekar A, Ray R. Ketamine dependence in an anesthesiologist: an occupational hazard? *Indian J Psychol Med.* 2014;36(3):335-337. doi:10.4103/0253-7176.135395 [PubMed 25035566]
97. Grabowski J, Goldin A, Arthur LG, et al. The effects of early anesthesia on neurodevelopment: a systematic review. *J Pediatr Surg.* 2021:S0022-3468(21)00014-2. doi:10.1016/j.jpedsurg.2021.01.002 [PubMed 33509654]
98. Graham MR. Clinical update regarding general anesthesia-associated neurotoxicity in infants and children. *Curr Opin Anaesthesiol.* 2017;30(6):682-687. doi:10.1097/ACO.0000000000000520 [PubMed 28915132]
99. Grant IS, Nimmo WS, Clements JA. Pharmacokinetics and analgesic effects of I.M. and oral ketamine. *Br J Anaesth.* 1981;53(8):805-810. doi:10.1093/bja/53.8.805 [PubMed 7272143]
100. Graudins A, Meek R, Egerton-Warburton D, Oakley E, Seith R. The PICHFORK (Pain in Children Fentanyl or Ketamine) trial: a randomized controlled trial comparing intranasal ketamine and

- fentanyl for the relief of moderate to severe pain in children with limb injuries. *Ann Emerg Med*. 2015;65(3):248-254.e1. doi:10.1016/j.annemergmed.2014.09.024 [PubMed 25447557]
101. Green SM, Roback MG, Kennedy RM, Krauss B. Clinical practice guideline for emergency department ketamine dissociative sedation: 2011 update. *Ann Emerg Med*. 2011;57(5):449-461. doi:10.1016/j.annemergmed.2010.11.030 [PubMed 21256625]
 102. Green SM, Roback MG, Krauss B, et al; Emergency Department Ketamine Meta-Analysis Study Group. Predictors of airway and respiratory adverse events with ketamine sedation in the emergency department: an individual-patient data meta-analysis of 8,282 children. *Ann Emerg Med*. 2009;54(2):158-68.e1-4. doi:10.1016/j.annemergmed.2008.12.011 [PubMed 19201064]
 103. Green SM, Sherwin TS. Incidence and severity of recovery agitation after ketamine sedation in young adults. *Am J Emerg Med*. 2005;23(2):142-144. doi:10.1016/j.ajem.2004.04.030 [PubMed 15765332]
 104. Greer A, Hewitt M, Khazaneh PT, et al. Ketamine versus etomidate for rapid sequence intubation: a systematic review and meta-analysis of randomized trials. *Crit Care Med*. 2025;53(2):e374-e383. doi:10.1097/CCM.0000000000006515 [PubMed 39570063]
 105. Groetzinger LM, Rivosecchi RM, Bain W, et al. Ketamine infusion for adjunct sedation in mechanically ventilated adults. *Pharmacotherapy*. 2018;38(2):181-188. doi:10.1002/phar.2065 [PubMed 29193185]
 106. Gropper MA, Eriksson LI, Fleisher LA, Wiener-Kronish JP, Cohen NH, Leslie K. *Miller's Anesthesia*. 9th ed. Elsevier; 2019.
 107. Guerra GG, Robertson CM, Alton GY, et al; Western Canadian Complex Pediatric Therapies Follow-up Group. Neurodevelopmental outcome following exposure to sedative and analgesic drugs for complex cardiac surgery in infancy. *Paediatr Anaesth*. 2011;21(9):932-941. doi:10.1111/j.1460-9592.2011.03581.x [PubMed 21507125]
 108. Guo J, Zhao F, Bian J, Hu Y, Tan J. Low-dose ketamine versus morphine in the treatment of acute pain in the emergency department: a meta-analysis of 15 randomized controlled trials. *Am J Emerg Med*. 2024;76:140-149. doi:10.1016/j.ajem.2023.11.056 [PubMed 38071883]
 109. Guthrie AM, Baum RA, Carter C, et al. Use of intranasal ketamine in pediatric patients in the emergency department. *Pediatr Emerg Care*. Published online July 8, 2019. doi:10.1097/PEC.0000000000001863 [PubMed 31290798]
 110. Gutstein HB, Johnson KL, Heard MB, Gregory GA. Oral ketamine preanesthetic medication in children. *Anesthesiology*. 1992;76(1):28-33. doi:10.1097/00000542-199201000-00004 [PubMed 1729932]
 111. Haas DA, Harper DG. Ketamine: a review of its pharmacologic properties and use in ambulatory anesthesia. *Anesth Prog*. 1992;39(3):61-68. [PubMed 1308374]
 112. Hall RW. Anesthesia and analgesia in the NICU. *Clin Perinatol*. 2012;39(1):239-254. [PubMed 22341549]
 113. Hatab SZ, Singh A, Felner EI, Kamat P. Transient central diabetes insipidus induced by ketamine infusion. *Ann Pharmacother*. 2014;48(12):1642-1645. doi:10.1177/1060028014549991 [PubMed 25225198]
 114. Heesen M, Böhmer J, Brinck EC, Kontinen VK, Klöhr S, Rossaint R, Straube S. Intravenous ketamine during spinal and general anaesthesia for caesarean section: systematic review and meta-analysis. *Acta Anaesthesiol Scand*. 2015;59(4):414-426. doi:10.1111/aas.12468. [PubMed 25789942]
 115. Hegenbarth MA; American Academy of Pediatrics Committee on Drugs. Preparing for pediatric emergencies: drugs to consider. *Pediatrics*. 2008;121(2):433-443. doi:10.1542/peds.2007-3284. [PubMed 18245435]

116. Heinrich M, Wetzstein V, Muensterer OJ, Till H. Conscious sedation: Off-label use of rectal S(+)-ketamine and midazolam for wound dressing changes in paediatric heat injuries. *Eur J Pediatr Surg*. 2004;14(4):235-239. doi:10.1055/s-2004-817960 [PubMed 15343462]
117. Heydari F, Gholamian A, Zamani M, Majidinejad S. Effect of intramuscular ketamine versus haloperidol on short-term control of severe agitated patients in emergency department; a randomized clinical trial. *Bull Emerg Trauma*. 2018;6(4):292-299. doi:10.29252/beat-060404 [PubMed 30402516]
118. Hocking G, Cousins MJ. Ketamine in chronic pain management: an evidence-based review. *Anesth Analg*. 2003;97(6):1730-1739. doi:10.1213/01.ANE.0000086618.28845.9B [PubMed 14633551]
119. Hugel V, Lauchart M, Magerl W, et al. Effects of low-dose intranasal (S)-ketamine in patients with neuropathic pain. *Eur J Pain*. 2010;14(4):387-394. doi:10.1016/j.ejpain.2009.08.002 [PubMed 19733106]
120. Ikonomidou C, Bosch F, Miksa M, et al. Blockade of NMDA receptors and apoptotic neurodegeneration in the developing brain. *Science*. 1999;283(5398):70-74. doi:10.1126/science.283.5398.70 [PubMed 9872743]
121. Iqbal O'Meara AM, Miller Ferguson N, Zven SE, et al. Potential neurodevelopmental effects of pediatric intensive care sedation and analgesia: repetitive benzodiazepine and opioid exposure alters expression of glial and synaptic proteins in juvenile rats. *Crit Care Explor*. 2020;2(4):e0105. doi:10.1097/CCE.000000000000105 [PubMed 32426747]
122. Isbister GK, Calver LA, Downes MA, Page CB. Ketamine as rescue treatment for difficult-to-sedate severe acute behavioral disturbance in the emergency department. *Ann Emerg Med*. 2016;67(5):581-587.e1. doi:10.1016/j.annemergmed.2015.11.028 [PubMed 26899459]
123. Ito S. Drug therapy for breast-feeding women. *N Engl J Med*. 2000;343(2):118-126. doi:10.1056/NEJM200007133430208 [PubMed 10891521]
124. Jabre P, Combes X, Lapostolle F, et al; KETASED Collaborative Study Group. Etomidate versus ketamine for rapid sequence intubation in acutely ill patients: a multicentre randomised controlled trial. *Lancet*. 2009;374(9686):293-300. doi:10.1016/S0140-6736(09)60949-1 [PubMed 19573904]
125. Jobeir A, Galal MO, Bulbul ZR, Solymar L, Darwish A, Schmaltz AA. Use of low-dose ketamine and/or midazolam for pediatric cardiac catheterization. *Pediatr Cardiol*. 2003;24(3):236-243. doi:10.1007/s00246-002-0339-5 [PubMed 12545318]
126. Kalsi SS, Wood DM, Dargan PI. The epidemiology and patterns of acute and chronic toxicity associated with recreational ketamine use. *Emerg Health Threats J*. 2011;4:7107. doi:10.3402/ehth.v4i0.7107 [PubMed 24149025]
127. Kannan TR, Saxena A, Bhatnagar S, Barry A. Oral ketamine as an adjuvant to oral morphine for neuropathic pain in cancer patients. *J Pain Symptom Manage*. 2002;23(1):60-65. doi:10.1016/s0885-3924(01)00373-6 [PubMed 11779670]
128. Ketalar (ketamine hydrochloride) [prescribing information]. Rochester, MI: Par Health USA; March 2026.
129. Ketalar (ketamine hydrochloride) [product monograph]. Montreal, Quebec, Canada: Searchlight Pharma Inc; May 2023.
130. Ketamine hydrochloride injection [prescribing information]. Lake Forest, IL: Hospira Inc; January 2013.
131. Khorsand SM. Maintenance of general anesthesia. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed September 21, 2020. .
132. Kitch BB. Out-of-hospital ketamine: review of a growing trend in patient care. *J Am Coll Emerg Physicians Open*. 2020;1(3):183-189. doi:10.1002/emp2.12023 [PubMed 33000033]

133. Köppel C, Arndt I, Ibe K. Effects of enzyme induction, renal and cardiac function on ketamine plasma kinetics in patients with ketamine long-term analgosedation. *Eur J Drug Metab Pharmacokinet.* 1990;15(3):259-263. doi:10.1007/BF03190213 [PubMed 2253657]
134. Koroki T, Kotani Y, Yaguchi T, et al. Ketamine versus etomidate as an induction agent for tracheal intubation in critically ill adults: a Bayesian meta-analysis. *Crit Care.* 2024;28(1):48. doi:10.1186/s13054-024-04831-4. [PubMed 38368326]
135. Koruk S, Mizrak A, Gul R, Kilic E, Yendi F, Oner U. Dexmedetomidine-ketamine and midazolam-ketamine combinations for sedation in pediatric patients undergoing extracorporeal shock wave lithotripsy: a randomized prospective study. *J Anesth.* 2010;24(6):858-863. doi:10.1007/s00540-010-1023-1. [PubMed 20924617]
136. Kramer AH. Early ketamine to treat refractory status epilepticus. *Neurocrit Care.* 2012;16(2):299-305. doi:10.1007/s12028-011-9668-7 [PubMed 22237581]
137. Kryst J, Kawalec P, Mitoraj AM, Pilc A, Lasoń W, Brzostek T. Efficacy of single and repeated administration of ketamine in unipolar and bipolar depression: a meta-analysis of randomized clinical trials. *Pharmacol Rep.* 2020;72(3):543-562. doi:10.1007/s43440-020-00097-z [PubMed 32301056]
138. Lago P, Garetti E, Merazzi D, et al. Guidelines for procedural pain in the newborn. *Acta Paediatr.* 2009;98(6):932-939. doi:10.1111/j.1651-2227.2009.01291.x [PubMed 19484828]
139. Lahti AC, Koffel B, LaPorte D, et al. Subanesthetic doses of ketamine stimulate psychosis in schizophrenia. *Neuropsychopharmacology.* 1995;13(1):9-19. doi:10.1016/0893-133X(94)00131-I [PubMed 8526975]
140. Lai Y, Wu S, Ni L, Chen Z, Li X, Yang S, Gui Y, Guan Z, Cai Z, Ye J. Ketamine-associated urinary tract dysfunction: an underrecognized clinical entity. *Urol Int.* 2012;89(1):93-96. doi:10.1159/000338098 [PubMed 22710265]
141. Legriel S, Oddo M, Brophy GM. What's new in refractory status epilepticus? *Intensive Care Med.* 2017;43(4):543-546. doi:10.1007/s00134-016-4501-6 [PubMed 27544138]
142. Li M, Martinelli AN, Oliver WD, Wilkerson RG. Evaluation of ketamine for excited delirium syndrome in the adult emergency department. *J Emerg Med.* 2019;S0736-S4679(19)30802-9. doi:10.1016/j.jemermed.2019.09.019 [PubMed 31735659]
143. Li Q, Wang S, Mei X. A single intravenous administration of a sub-anesthetic ketamine dose during the perioperative period of cesarean section for preventing postpartum depression: a meta-analysis. *Psychiatry Res.* 2022;310:114396. doi:10.1016/j.psychres.2022.114396 [PubMed 35278826]
144. Li S, Zhou W, Li P, Lin R. Effects of ketamine and esketamine on preventing postpartum depression after cesarean delivery: a meta-analysis. *J Affect Disord.* 2024;351:720-728. doi:10.1016/j.jad.2024.01.202 [PubMed 38286233]
145. Lim DK. Ketamine associated psychedelic effects and dependence. *Singapore Med J.* 2003;44(1):31-34. [PubMed 12762561]
146. Lin C, Durieux ME. Ketamine and kids: an update. *Paediatr Anaesth.* 2005;15(2):91-97. doi:10.1111/j.1460-9592.2005.01475.x [PubMed 15675923]
147. Lin PC, Lane HY, Lin CH. Spontaneous remission of ketamine withdrawal-related depression. *Clin Neuropharmacol.* 2016;39(1):51-52. doi:10.1097/WNF.0000000000000121 [PubMed 26757313]
148. Lin SM, Liu K, Tsai SK, Lee TY. Rectal ketamine versus intranasal ketamine as premedicant in children. *Ma Zui Xue Za Zhi.* 1990;28(2):177-183. [PubMed 2215104]
149. Linder LM, Ross CA, Weant KA. Ketamine for the acute management of excited delirium and agitation in the prehospital setting. *Pharmacotherapy.* 2018;38(1):139-151. doi:10.1002/phar.2060 [PubMed 29136301]

150. List WF, Crumrine RS, Cascorbi HF, Weiss MH. Increased cerebrospinal fluid pressure after ketamine. *Anesthesiology*. 1972;36(1):98-99. doi:10.1097/00000542-197201000-00023 [PubMed 5006995]
151. Little B, Chang T, Chucot L, et al. Study of ketamine as an obstetric anesthetic agent. *Am J Obstet Gynecol*. 1972;15:113(2):247-260. doi:10.1016/0002-9378(72)90774-0 [PubMed 4554554]
152. Loo CK, Gálvez V, O'Keefe E, et al. Placebo-controlled pilot trial testing dose titration and intravenous, intramuscular and subcutaneous routes for ketamine in depression. *Acta Psychiatr Scand*. 2016;134(1):48-56. doi:10.1111/acps.12572 [PubMed 27028832]
153. Louon A, Reddy VG. Nasal midazolam and ketamine for paediatric sedation during computerised tomography. *Acta Anaesthesiol Scand*. 1994;38(3):259-261. doi:10.1111/j.1399-6576.1994.tb03885.x [PubMed 8023666]
154. Loveday BA, Sindt J. Ketamine protocol for palliative care in cancer patients with refractory pain. *J Adv Pract Oncol*. 2015;6(6):555-561. [PubMed 27648345]
155. Lovett S, Reed T, Riggs R, et al. A randomized, noninferiority, controlled trial of two doses of intravenous subdissociative ketamine for analgesia in the emergency department. *Acad Emerg Med*. 2021;28(6):647-654. doi:10.1111/acem.14200 [PubMed 33354815]
156. Lyon F, Dabbs T, O'Meara M. Ketamine sedation during the treatment of retinopathy of prematurity. *Eye (Lond)*. 2008;22(5):684-686. doi:10.1038/sj.eye.6702717 [PubMed 17417623]
157. MacQueen GM, Frey BN, Ismail Z, et al. Canadian Network for Mood and Anxiety Treatments (CANMAT) 2016 clinical guidelines for the management of adults with major depressive disorder: section 6. Special populations: youth, women, and the elderly. *Can J Psychiatry*. 2016;61(9):588-603. doi:10.1177/0706743716659276 [PubMed 27486149]
158. Majdinasab E, Datta P, Krutsch K, Baker T, Hale TW. Pharmacokinetics of ketamine transfer into human milk. *J Clin Psychopharmacol*. 2023;43(5):407-410. doi:10.1097/JCP.0000000000001711 [PubMed 37683228]
159. Malhotra AK, Pinals DA, Adler CM, et al. Ketamine-induced exacerbation of psychotic symptoms and cognitive impairment in neuroleptic-free schizophrenics. *Neuropsychopharmacology*. 1997;17(3):141-150. doi:10.1016/S0893-133X(97)00036-5 [PubMed 9272481]
160. Malinovsky JM, Servin F, Cozian A, Lepage JY, Pinaud M. Ketamine and norketamine plasma concentrations after i.v., nasal and rectal administration in children. *Br J Anaesth*. 1996;77(2):203-207. doi:10.1093/bja/77.2.203 [PubMed 8881626]
161. Marchetti F, Coutaux A, Bellanger A, Magneux C, Bourgeois P, Mion G. Efficacy and safety of oral ketamine for the relief of intractable chronic pain: A retrospective 5-year study of 51 patients. *Eur J Pain*. 2015;19(7):984-993. doi:10.1002/ejp.624 [PubMed 25381898]
162. Martin E, Vickers B, Landau R, Reece-Stremtan S. ABM clinical protocol #28, peripartum analgesia and anesthesia for the breastfeeding mother. *Breastfeed Med*. 2018;13(3):164-171. doi:10.1089/bfm.2018.29087.ejm [PubMed 29595994]
163. McAllister-Williams RH, Baldwin DS, Cantwell R, et al; endorsed by the British Association for Psychopharmacology. British Association for Psychopharmacology consensus guidance on the use of psychotropic medication preconception, in pregnancy and postpartum 2017. *J Psychopharmacol*. 2017;31(5):519-552. doi:10.1177/0269881117699361 [PubMed 28440103]
164. McGirr A, Berlim MT, Bond DJ, Fleck MP, Yatham LN, Lam RW. A systematic review and meta-analysis of randomized, double-blind, placebo-controlled trials of ketamine in the rapid treatment of major depressive episodes. *Psychol Med*. 2015;45(4):693-704. doi:10.1017/S0033291714001603 [PubMed 25010396]
165. McGlone RG, Howes MC, Joshi M. The Lancaster Experience of 2.0 to 2.5 mg/kg Intramuscular Ketamine for Paediatric Sedation: 501 Cases and Analysis. *Emerg Med J*. 2004;21(3):290-295. doi:10.1136/emj.2002.003772 [PubMed 15107365]

166. McPherson C. Sedation and analgesia in mechanically ventilated preterm neonates: continue standard of care or experiment? *J Pediatr Pharmacol Ther.* 2012;17(4):351-364. doi:10.5863/1551-6776-17.4.351 [PubMed 23413121]
167. Melendez E, Bachur R. Serious adverse events during procedural sedation with ketamine. *Pediatr Emerg Care.* 2009;25(5):325-328. doi:10.1097/PEC.0b013e3181a341e0 [PubMed 19404223]
168. Mellon RD, Simone AF, Rappaport BA. Use of anesthetic agents in neonates and young children. *Anesth Analg.* 2007;104(3):509-520. doi:10.1213/01.ane.0000255729.96438.b0 [PubMed 17312200]
169. Mercadante S, Arcuri E, Tirelli W, Casuccio A. Analgesic effect of intravenous ketamine in cancer patients on morphine therapy: a randomized, controlled, double-blind, crossover, double-dose study. *J Pain Symptom Manage.* 2000;20(4):246-252. doi:10.1016/s0885-3924(00)00194-9 [PubMed 11027905]
170. Mercadante S, Caruselli A, Casuccio A. The use of ketamine in a palliative-supportive care unit: a retrospective analysis. *Ann Palliat Med.* 2018;7(2):205-210. doi:10.21037/apm.2018.01.01 [PubMed 29764182]
171. Merelman AH, Perlmutter MC, Strayer RJ. Alternatives to rapid sequence intubation: contemporary airway management with ketamine. *West J Emerg Med.* 2019;20(3):466-471. doi:10.5811/westjem.2019.4.42753 [PubMed 31123547]
172. Mikellides G, Michael P, Psalta L, Schuhmann T, Sack AT. A retrospective naturalistic study comparing the efficacy of ketamine and repetitive transcranial magnetic stimulation for treatment-resistant depression. *Front Psychiatry.* 2022;12:784830. doi:10.3389/fpsy.2021.784830 [PubMed 35095600]
173. Miller M, Kruit N, Heldreich C, et al. Hemodynamic response after rapid sequence induction with ketamine in out-of-hospital patients at risk of shock as defined by the shock index. *Ann Emerg Med.* 2016;68(2):181-188.e2. doi:10.1016/j.annemergmed.2016.03.041 [PubMed 27130803]
174. Miller RD. *Miller's Anesthesia.* 7th ed. Churchill Livingstone; 2010.
175. Mion G, Villevieille T. Ketamine pharmacology: an update (pharmacodynamics and molecular aspects, recent findings). *CNS Neurosci Ther.* 2013;19(6):370-380. doi:10.1111/cns.12099 [PubMed 23575437]
176. Moaddel R, Abdrakhmanova G, Kozak J, et al. Sub-anesthetic concentrations of (R,S)-ketamine metabolites inhibit acetylcholine-evoked currents in $\alpha 7$ nicotinic acetylcholine receptors. *Eur J Pharmacol.* 2013;698(1-3):228-234. doi:10.1016/j.ejphar.2012.11.023 [PubMed 23183107]
177. Moore MJ, Im D. The acutely agitated or violent adult: pharmacologic management. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <https://www.uptodate.com>. Accessed April 15, 2025.
178. Moore NN, Bostwick JM. Ketamine dependence in anesthesia providers. *Psychosomatics.* 1999;40(4):356-359. doi:10.1016/S0033-3182(99)71231-4 [PubMed 10402883]
179. Mosier JM, Sakles JC. The physiologically difficult airway: optimization of airway management in adults at risk of decompensation for emergency medicine and critical care. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <https://www.uptodate.com>. Accessed May 5, 2025.
180. Murrough JW, Perez AM, Pillemer S, et al. Rapid and longer-term antidepressant effects of repeated ketamine infusions in treatment-resistant major depression. *Biol Psychiatry.* 2013;74(4):250-256. doi:10.1016/j.biopsych.2012.06.022 [PubMed 22840761]
181. Nagdeve NG, Yaddanapudi S, Pandav SS. The effect of different doses of ketamine on intraocular pressure in anesthetized children. *J Pediatr Ophthalmol Strabismus.* 2006;43(4):219-223. doi:10.3928/01913913-20060701-03 [PubMed 16915900]

182. Nazarian DJ, Broder JS, Thiessen MEW, Wilson MP, Zun LS, Brown MD; American College of Emergency Physicians Clinical Policies Subcommittee (Writing Committee) on the Adult Psychiatric Patient. Clinical policy: critical issues in the diagnosis and management of the adult psychiatric patient in the emergency department. *Ann Emerg Med.* 2017;69(4):480-498. doi:10.1016/j.annemergmed.2017.01.036 [PubMed 28335913]
183. Nemecek BD, Hammond DA. *Demystifying Drug Dosing in Renal Dysfunction*. American Society of Health-System Pharmacists; 2019.
184. Neuman G, Koren G. Safety of procedural sedation in pregnancy. *J Obstet Gynaecol Can.* 2013;35(2):168-173. doi:10.1016/S1701-2163(15)31023-9. [PubMed 23470068]
185. Newport DJ, Carpenter LL, McDonald WM, Potash JB, Tohen M, Nemeroff CB; APA Council of Research Task Force on Novel Biomarkers and Treatments. Ketamine and other NMDA antagonists: early clinical trials and possible mechanisms in depression. *Am J Psychiatry.* 2015;172(10):950-966. doi:10.1176/appi.ajp.2015.15040465 [PubMed 26423481]
186. Nielsen BN, Friis SM, Rømsing J, et al. Intranasal sufentanil/ketamine analgesia in children. *Paediatr Anaesth.* 2014;24(2):170-180. doi:10.1111/pan.12268 [PubMed 24118506]
187. Niesters M, Martini C, Dahan A. Ketamine for chronic pain: Risks and benefits. *Br J Clin Pharmacology.* 2013;77(2):357-367. doi:10.1111/bcp.12094
188. Norambuena C, Yañez J, Flores V, Puentes P, Carrasco P, Villena R. Oral ketamine and midazolam for pediatric burn patients: a prospective, randomized, double-blind study. *J Pediatr Surg.* 2013;48(3):629-634. doi:10.1016/j.jpedsurg.2012.08.018 [PubMed 23480923]
189. O'Brien ME, Fuh L, Raja AS, White BA, Yun BJ, Hayes BD. Reduced-dose intramuscular ketamine for severe agitation in an academic emergency department. *Clin Toxicol (Phila).* 2020;58(4):294-298. doi:10.1080/15563650.2019.1643468 [PubMed 31335216]
190. Oddo M, Bracard S, Cariou A, et al. Update in neurocritical care: a summary of the 2018 Paris international conference of the French Society of Intensive Care. *Ann Intensive Care.* 2019;9(1):47. doi:10.1186/s13613-019-0523-x [PubMed 30993550]
191. Okon T. Ketamine: an introduction for the pain and palliative medicine physician. *Pain Physician.* 2007;10(3):493-500. [PubMed 17525784]
192. Oliveira J E Silva L, Lee JY, Bellolio F, Homme JL, Anderson JL. Intranasal ketamine for acute pain management in children: A systematic review and meta-analysis. *Am J Emerg Med.* 2020;38(9):1860-1866. doi:10.1016/j.ajem.2020.05.094 [PubMed 32739857]
193. Olutoye OA, Baker BW, Belfort MA, Olutoye OO. Food and Drug Administration warning on anesthesia and brain development: implications for obstetric and fetal surgery. *Am J Obstet Gynecol.* 2018;218(1):98-102. doi:10.1016/j.ajog.2017.08.107 [PubMed 28888583]
194. O'Mahony D, Cherubini A, Guiteras AR, et al. STOPP/START criteria for potentially inappropriate prescribing in older people: version 3. *Eur Geriatr Med.* 2023;14(4):625-632. doi:10.1007/s41999-023-00777-y [PubMed 37256475]
195. Orhurhu V, Orhurhu MS, Bhatia A, Cohen SP. Ketamine infusions for chronic pain: a systematic review and meta-analysis of randomized controlled trials. *Anesth Analg.* 2019;129(1):241-254. doi:10.1213/ANE.0000000000004185 [PubMed 31082965]
196. Ozdemir D, Kayserili E, Arslanoglu S, Gulez P, Vergin C. Ketamine and midazolam for invasive procedures in children with malignancy: a comparison of routes of intravenous, oral, and rectal administration. *J Trop Pediatr.* 2004;50(4):224-228. doi:10.1093/tropej/50.4.224 [PubMed 15357562]
197. Page RL 2nd, O'Bryant CL, Cheng D, et al; American Heart Association Clinical Pharmacology and Heart Failure and Transplantation Committees of the Council on Clinical Cardiology; Council on Cardiovascular Surgery and Anesthesia; Council on Cardiovascular and Stroke Nursing; and Council on Quality of Care and Outcomes Research. Drugs That May Cause or Exacerbate Heart Failure: A Scientific Statement From the American Heart Association [published

- correction appears in *Circulation*. 2016;134(12):e261]. *Circulation*. 2016;134(6):e32-e69. doi:10.1161/CIR.0000000000000426 [PubMed 27400984]
198. Pai A, Heining M. Ketamine. Continuing education in anaesthesia, critical care and pain. *Br J Anaesthesia*. 2007;7(2):59-63. Accessed December 2021. <https://academic.oup.com/bjaed/article/7/2/59/3>
 199. Pal HR, Berry N, Kumar R, Ray R. Ketamine dependence. *Anaesth Intensive Care*. 2002;30(3):382-384. doi:10.1177/0310057X0203000323 [PubMed 12075653]
 200. Pal R, Balt S, Erowid E, et al. Ketamine is associated with lower urinary tract signs and symptoms. *Drug Alcohol Depend*. 2013;132(1-2):189-194. doi:10.1016/j.drugalcdep.2013.02.005 [PubMed 23474358]
 201. Patel P, Sun L. Update on neonatal anesthetic neurotoxicity: insight into molecular mechanisms and relevance to humans. *Anesthesiology*. 2009;110(4):703-708. doi:10.1097/ALN.0b013e31819c42a4 [PubMed 19276968]
 202. Pees C, Haas NA, Ewert P, Berger F, Lange PE. Comparison of analgesic/sedative effect of racemic ketamine and S(+)-ketamine during cardiac catheterization in newborns and children. *Pediatr Cardiol*. 2003;24(5):424-429. doi:10.1007/s00246-002-0356-4 [PubMed 14627307]
 203. Peng TR, Lee MC, Wu TW, Lan CC. Suspected ketamine-associated lower urinary tract symptoms. *Urol J*. 2014;11(2):1508-1510. [PubMed 24807773]
 204. Phillips JL, Norris S, Talbot J, et al. Single and repeated ketamine infusions for reduction of suicidal ideation in treatment-resistant depression. *Neuropsychopharmacology*. 2020;45(4):606-612. doi:10.1038/s41386-019-0570-x [PubMed 31759333]
 205. Phillips JL, Norris S, Talbot J, et al. Single, repeated, and maintenance ketamine infusions for treatment-resistant depression: a randomized controlled trial. *Am J Psychiatry*. 2019;176(5):401-409. doi:10.1176/appi.ajp.2018.18070834 [PubMed 30922101]
 206. Poonai N, Canton K, Ali S, et al. Intranasal ketamine for procedural sedation and analgesia in children: A systematic review. *PLoS One*. 2017;12(3):e0173253. doi:10.1371/journal.pone.0173253 [PubMed 28319161]
 207. Potnuru P. Awake tracheal intubation. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed August 27, 2024.
 208. Powers AR 3rd, Gancsos MG, Finn ES, Morgan PT, Corlett PR. Ketamine-induced hallucinations. *Psychopathology*. 2015;48(6):376-385. doi:10.1159/000438675 [PubMed 26361209]
 209. Quibell R, Fallon M, Mihalyo M, Twycross R, Wilcock A. Ketamine. *J Pain Symptom Manage*. 2015;50(2):268-278. doi:10.1016/j.jpainsymman.2015.06.002 [PubMed 26096492]
 210. Quibell R, Prommer EE, Mihalyo M, Twycross R, Wilcock A. Ketamine*. *J Pain Symptom Manage*. 2011;41(3):640-649. doi:10.1016/j.jpainsymman.2011.01.001 [PubMed 21419322]
 211. Rai S, Drislane FW. Treatment of refractory and super-refractory status epilepticus. *Neurotherapeutics*. 2018;15(3):697-712. doi:10.1007/s13311-018-0640-5 [PubMed 29922905]
 212. Rasmussen KG, Lineberry TW, Galardy CW, et al. Serial infusions of low-dose ketamine for major depression. *J Psychopharmacol*. 2013;27(5):444-450. doi:10.1177/0269881113478283. [PubMed 23428794]
 213. Rech MA, Barbas B, Chaney W, Greenhalgh E, Turck C. When to pick the nose: out-of-hospital and emergency department intranasal administration of medications. *Ann Emerg Med*. 2017;70(2):203-211. doi:10.1016/j.annemergmed.2017.02.015 [PubMed 28366351]
 214. Reece-Stremtan S, Campos M, Kokajko L; Academy of Breastfeeding Medicine. ABM clinical protocol #15: analgesia and anesthesia for the breastfeeding mother, revised 2017. *Breastfeed Med*. 2017;12(9):500-506. doi:10.1089/bfm.2017.29054.srt [PubMed 29624435]
 215. Refer to manufacturer's labeling.

216. Reynolds SL, Bryant KK, Studnek JR, et al. Randomized controlled feasibility trial of intranasal ketamine compared to intranasal fentanyl for analgesia in children with suspected extremity fractures. *Acad Emerg Med*. 2017;24(12):1430-1440. doi:10.1111/acem.13313 [PubMed 28926159]
217. Riddell J, Tran A, Bengiamin R, Hendey GW, Armenian P. Ketamine as a first-line treatment for severely agitated emergency department patients. *Am J Emerg Med*. 2017;35(7):1000-1004. doi:10.1016/j.ajem.2017.02.026 [PubMed 28237385]
218. Roberts JR, Custalow CB, Thomsen TW, et al, eds. *Roberts and Hedges' Clinical Procedures in Emergency Medicine and Acute Care*. 7th ed. Elsevier. 2019.
219. Rock MJ, Reyes de la Rocha S, L'Hommedieu CS, Truemper E. Use of ketamine in asthmatic children to treat respiratory failure refractory to conventional therapy. *Crit Care Med*. 1986;14(5):514-516. [PubMed 3698618]
220. Roelofse JA, Shipton EA, de la Harpe CJ, Blignaut RJ. Intranasal sufentanil/midazolam versus ketamine/midazolam for analgesia/sedation in the pediatric population prior to undergoing multiple dental extractions under general anesthesia: a prospective, double-blind, randomized comparison. *Anesth Prog*. 2004;51(4):114-121. [PubMed 15675259]
221. Rosati A, De Masi S, Guerrini R. Ketamine for refractory status epilepticus: a systematic review. *CNS Drugs*. 2018;32(11):997-1009. doi:10.1007/s40263-018-0569-6 [PubMed 30232735]
222. Sanacora G, Frye MA, McDonald W, et al; American Psychiatric Association (APA) Council of Research Task Force on Novel Biomarkers and Treatments. A consensus statement on the use of ketamine in the treatment of mood disorders. *JAMA Psychiatry*. 2017;74(4):399-405. doi:10.1001/jamapsychiatry.2017.0080 [PubMed 28249076]
223. Schep LJ, Slaughter RJ, Watts M, Mackenzie E, Gee P. The clinical toxicology of ketamine. *Clin Toxicol (Phila)*. 2023;61(6):415-428. doi:10.1080/15563650.2023.2212125 [PubMed 37267048]
224. Schwenk ES, Viscusi ER, Buvanendran A, et al. Consensus guidelines on the use of intravenous ketamine infusions for acute pain management from the American Society of Regional Anesthesia and Pain Medicine, the American Academy of Pain Medicine, and the American Society of Anesthesiologists. *Reg Anesth Pain Med*. 2018;43(5):456-466. doi:10.1097/AAP.0000000000000806 [PubMed 29870457]
225. Shah A, Mosdosy G, McLeod S, Lehnhardt K, Peddle M, Rieder M. A blinded, randomized controlled trial to evaluate ketamine/propofol versus ketamine alone for procedural sedation in children. *Ann Emerg Med*. 2011;57(5):425-433.e2. doi:10.1016/j.annemergmed.2010.08.032 [PubMed 20947210]
226. Shahane A, Zhao W, Pakalapati N, et al. Simultaneous quantitation of ketamine, nor-ketamine and dehydronorketamine in human milk using a novel ultra high-performance liquid chromatography-mass spectrometry (UPLC-MS/MS) assay. *J Pharm Biomed Anal*. 2023;234:115502. doi:10.1016/j.jpba.2023.115502 [PubMed 37311376]
227. Shimonovich S, Gigi R, Shapira A, et al. Intranasal ketamine for acute traumatic pain in the emergency department: a prospective, randomized clinical trial of efficacy and safety. *BMC Emerg Med*. 2016;16(1):43. doi:10.1186/s12873-016-0107-0 [PubMed 27829367]
228. Shiroma PR, Johns B, Kuskowski M, et al. Augmentation of response and remission to serial intravenous subanesthetic ketamine in treatment resistant depression. *J Affect Disord*. 2014;155:123-129. doi:10.1016/j.jad.2013.10.036 [PubMed 24268616]
229. Shrestha R, Pant S, Shrestha A, Batajoo KH, Thapa R, Vaidya S. Intranasal ketamine for the treatment of patients with acute pain in the emergency department. *World J Emerg Med*. 2016;7(1):19-24. doi:10.5847/wjem.j.1920-8642.2016.01.003 [PubMed 27006733]
230. Singh JB, Fedgchin M, Daly EJ, et al. A double-blind, randomized, placebo-controlled, dose-frequency study of intravenous ketamine in patients with treatment-resistant depression. *Am J Psychiatry*. 2016;173(8):816-826. doi:10.1176/appi.ajp.2016.16010037 [PubMed 27056608]

231. Singh V, Gillespie TW, Harvey RD. Intranasal ketamine and its potential role in cancer-related pain. *Pharmacotherapy*. 2018;38(3):390-401. doi:10.1002/phar.2090 [PubMed 29396996]
232. Soto E, Stewart DR, Mannes AJ, et al. Oral ketamine in the palliative care setting: a review of the literature and case report of a patient with neurofibromatosis type 1 and glomus tumor-associated complex regional pain syndrome. *Am J Hosp Palliat Care*. 2012;29(4): 308-317. doi:10.1177/1049909111416345 [PubMed 21803784]
233. Society for Pediatric Sedation (SPS). FDA advisory on anesthesia and sedation medication in children. Society for Pediatric Sedation website. [https://www.pedsedation.org/resources/quality-safety/fda-advisory/#:~:text=In%20December%202016%2C%20the%20FDA,\(http%3A%2F%2Fwww.fda.](https://www.pedsedation.org/resources/quality-safety/fda-advisory/#:~:text=In%20December%202016%2C%20the%20FDA,(http%3A%2F%2Fwww.fda.) Accessed April 2021.
234. Srinivas C, Tiwari T, Prakash R, Prabha R, Raman R, Arshad Z. A randomized comparison of intravenous versus nebulized routes for administering dexmedetomidine and ketamine combination to facilitate awake fiberoptic intubation. *Cureus*. 2023;15(4):e38322. doi:10.7759/cureus.38322 [PubMed 37261179]
235. Sriraman NK, Melvin K, Meltzer-Brody S. ABM clinical protocol #18: use of antidepressants in breastfeeding mothers. *Breastfeed Med*. 2015;10(6):290-299. doi:10.1089/bfm.2015.29002 [PubMed 26204124]
236. Stollings JL, Diedrich DA, Oyen LJ, Brown DR. Rapid-sequence intubation: a review of the process and considerations when choosing medications. *Ann Pharmacother*. 2014;48(1):62-76. doi:10.1177/1060028013510488 [PubMed 24259635]
237. Su PH, Chang YZ, Chen JY. Infant with in utero ketamine exposure: quantitative measurement of residual dosage in hair. *Pediatr Neonatol*. 2010;51(5):279-284. doi:10.1016/S1875-9572(10)60054-X [PubMed 20951358]
238. Sungur Ulke Z, Kartal U, Orhan Sungur M, Camci E, Tugrul M. Comparison of sevoflurane and ketamine for anesthetic induction in children with congenital heart disease. *Paediatr Anaesth*. 2008;18(8):715-21. doi:10.1111/j.1460-9592.2008.02637.x. [PubMed 18544145]
239. Surendar MN, Pandey RK, Saksena AK, Kumar R, Chandra G. A comparative evaluation of intranasal dexmedetomidine, midazolam and ketamine for their sedative and analgesic properties: a triple blind randomized study. *J Clin Pediatr Dent*. 2014;38(3):255-261. doi:10.17796/jcpd.38.3.l828585807482966 [PubMed 25095322]
240. Swainson J, McGirr A, Blier P, et al. The Canadian Network for Mood and Anxiety Treatments (CANMAT) task force recommendations for the use of racemic ketamine in adults with major depressive disorder: recommendations Du Groupe De Travail Du Réseau Canadien Pour Les Traitements De L'humeur Et De L'anxiété (Canmat) Concernant L'utilisation De La Kétamine Racémique Chez Les Adultes Souffrant De Trouble Dépressif Majeur. *Can J Psychiatry*. 2021;66(2):113-125. doi:10.1177/0706743720970860 [PubMed 33174760]
241. Szarmach J, Cubała WJ, Włodarczyk A, Wiglusz MS. Short-term ketamine administration in treatment-resistant depression: focus on cardiovascular safety. *Psychiatr Danub*. 2019;31(suppl 3):585-590. [PubMed 31488795]
242. Talahma M, Sabharwal V, Bukovskaya Y, Khan F. Ketamine infusion used to successfully control refractory status epilepticus in a pregnant patient. *Case Rep Neurol Med*. 2018;2018:3041279. doi:10.1155/2018/3041279 [PubMed 30498609]
243. Tanaka M, Sato M, Saito A, Nishikawa T. Reevaluation of rectal ketamine premedication in children: comparison with rectal midazolam. *Anesthesiology*. 2000;93(5):1217-1224. doi:10.1097/00000542-200011000-00014 [PubMed 11046209]
244. Tawfic QA, Faris AS, Kausalya R. The role of a low-dose ketamine-midazolam regimen in the management of severe painful crisis in patients with sickle cell disease. *J Pain Symptom Manage*. 2014;47(2):334-340. doi:10.1016/j.jpainsymman.2013.03.012 [PubMed 23856095]

245. Thase M, Connolly KR. Ketamine and esketamine for treating unipolar depression in adults: administration and adverse effects. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed January 6, 2026.
246. Tobias JD, Martin LD, Wetzel RC. Ketamine by continuous infusion for sedation in the pediatric intensive care unit. *Crit Care Med*. 1990;18(8):819-821. doi:10.1097/00003246-199008000-00004 [PubMed 2379394]
247. Tobias JD, Phipps S, Smith B, Mulhern RK. Oral ketamine premedication to alleviate the distress of invasive procedures in pediatric oncology patients. *Pediatrics*. 1992;90(4):537-541. [PubMed 1408506]
248. Tobias JD, Rasmussen GE. Pain management and sedation in the pediatric intensive care unit. *Pediatr Clin North Am*. 1994;41(6):1269-1292. doi:10.1016/s0031-3955(16)38873-3 [PubMed 7984386]
249. Tomatir E, Atalay H, Gurses E, Erbay H, Bozkurt P. Effects of low dose ketamine before induction on propofol anesthesia for pediatric magnetic resonance imaging. *Paediatr Anaesth*. 2004;14(10):845-850. doi:10.1111/j.1460-9592.2004.01303.x [PubMed 15385013]
250. Tsubo T, Sakai I, Okawa H, Ishihara H, Matsuki A. Ketamine and midazolam kinetics during continuous hemodiafiltration in patients with multiple organ dysfunction syndrome. *Intensive Care Med*. 2001;27(6):1087-1090. doi:10.1007/s001340100965 [PubMed 11497143]
251. Turhanoglu S, Kararmaz A, Ozyilmaz MA, Kaya S, Tok D. Effects of different doses of oral ketamine for premedication of children. *Eur J Anaesthesiol*. 2003;20(1):56-60. doi:10.1017/s0265021503000103 [PubMed 12553389]
252. Udy AA, Roberts JA, Boots RJ, Paterson DL, Lipman J. Augmented renal clearance: implications for antibacterial dosing in the critically ill. *Clin Pharmacokinet*. 2010;49(1):1-16. doi:10.2165/11318140-000000000-00000 [PubMed 20000886]
253. Upchurch CP, Grijalva CG, Russ S, et al. Comparison of etomidate and ketamine for induction during rapid sequence intubation of adult trauma patients. *Ann Emerg Med*. 2017;69(1):24-33.e2. doi:10.1016/j.annemergmed.2016.08.009 [PubMed 27993308]
254. US Department of Veterans Affairs/Department of Defense (VA/DoD). VA/DoD clinical practice guideline for the management of major depressive disorder. <https://www.healthquality.va.gov/guidelines/MH> Updated February 2022. Accessed June 15, 2022.
255. Vadivelu N, Schermer E, Kodumudi V, Belani K, Urman RD, Kaye AD. Role of ketamine for analgesia in adults and children. *J Anaesthesiol Clin Pharmacol*. 2016;32(3):298-306. doi:10.4103/0970-9185.168149 [PubMed 27625475]
256. van der Bijl P, Roelofse JA, Stander IA. Rectal ketamine and midazolam for premedication in pediatric dentistry. *J Oral Maxillofac Surg*. 1991;49(10):1050-1054. doi:10.1016/0278-2391(91)90136-a [PubMed 1890517]
257. Vande Voort JL, Morgan RJ, Kung S, et al. Continuation phase intravenous ketamine in adults with treatment-resistant depression. *J Affect Disord*. 2016;206:300-304. doi:10.1016/j.jad.2016.09.008 [PubMed 27656788]
258. Verma AK, Verma S, Barik AK, Kanaujia V, Arya S. Intubating conditions and hemodynamic changes during awake fiberoptic intubation using fentanyl with ketamine versus dexmedetomidine for anticipated difficult airway: a randomized clinical trial. *Braz J Anesthesiol*. 2021;71(3):259-264. doi:10.1016/j.bjane.2021.01.005 [PubMed 33744331]
259. Wagner BK, O'Hara DA. Pharmacokinetics and pharmacodynamics of sedatives and analgesics in the treatment of agitated critically ill patients. *Clin Pharmacokinet*. 1997;33(6):426-453. doi:10.2165/00003088-199733060-00003 [PubMed 9435992]
260. Wang J, Echevarria GC, Doan L, et al. Effects of a single subanaesthetic dose of ketamine on pain and mood after laparoscopic bariatric surgery: a randomised double-blind placebo controlled

- study. *Eur J Anaesthesiol*. 2019;36(1):16-24. doi:10.1097/EJA.0000000000000860 [PubMed 30095550]
261. Wang J, Xu Z, Feng Z, Ma R, Zhang X. Impact of ketamine on pain management in cesarean section: a systematic review and meta-analysis. *Pain Physician*. 2020;23(2):135-148. [PubMed 32214291]
 262. Wang X, Zhou ZJ, Zhang XF, Zheng S. A comparison of two different doses of rectal ketamine added to 0.5 mg x kg(-1) midazolam and 0.02 mg x kg(-1) atropine in infants and young children. *Anaesth Intensive Care*. 2010;38(5):900-904. doi:10.1177/0310057X1003800515 [PubMed 20865876]
 263. Webster LR, Walker MJ. Safety and efficacy of prolonged outpatient ketamine infusions for neuropathic pain. *Am J Ther*. 2006;13(4):300-305. doi:10.1097/00045391-200607000-00004 [PubMed 16858163]
 264. Wei YB, Yang JR, Yin Z, Guo Q, Liang BL, Zhou KQ. Genitourinary toxicity of ketamine. *Hong Kong Med J*. 2013;19(4):341-348. doi:10.12809/hkmj134013 [PubMed 23832948]
 265. Weiss M, Siegmund W. Pharmacokinetic modeling of ketamine enantiomers and their metabolites after administration of prolonged-release ketamine with emphasis on 2,6-hydroxynorketamines. *Clin Pharmacol Drug Dev*. 2022;11(2):194-206. doi:10.1002/cpdd.993 [PubMed 34265182]
 266. Weksler N, Ovadia L, Muati G, Stav A. Nasal ketamine for paediatric premedication. *Can J Anaesth*. 1993;40(2):119-121. doi:10.1007/BF03011307 [PubMed 8443849]
 267. White PF, Way WL, Trevor AJ. Ketamine – its pharmacology and therapeutic uses. *Anesthesiology*. 1982;56(2):119-136. doi:10.1097/0000542-198202000-00007 [PubMed 6892475]
 268. Willman EV, Andolfatto G. A prospective evaluation of "ketofol" (ketamine/propofol combination) for procedural sedation and analgesia in the emergency department. *Ann Emerg Med*. 2007;49(1):23-30. doi:10.1016/j.annemergmed.2006.08.002 [PubMed 17059854]
 269. Witt K, Potts J, Hubers A, et al. Ketamine for suicidal ideation in adults with psychiatric disorders: a systematic review and meta-analysis of treatment trials. *Aust N Z J Psychiatry*. 2020;54(1):29-45. doi:10.1177/0004867419883341 [PubMed 31729893]
 270. Wolfson P, Cole R, Lynch K, et al. The pharmacokinetics of ketamine in the breast milk of lactating women: quantification of ketamine and metabolites. *J Psychoactive Drugs*. 2023;55(3):354-358. doi:10.1080/02791072.2022.2101903 [PubMed 35880962]
 271. Wu L, Lalwani K, Hook KA, Almarino BM, Fu R, Edmunds B. Respiratory complications associated with ketamine anesthesia for ophthalmic procedures following intraocular pressure measurement in children. *J Anaesthesiol Clin Pharmacol*. 2014;30(2):253-257. doi:10.4103/0970-9185.130047 [PubMed 24803768]
 272. Xiong J, Lipsitz O, Chen-Li D, et al. The acute antisuicidal effects of single-dose intravenous ketamine and intranasal esketamine in individuals with major depression and bipolar disorders: a systematic review and meta-analysis. *J Psychiatr Res*. 2021;134:57-68. doi:10.1016/j.jpsychires.2020.12.038 [PubMed 33360864]
 273. Yan J, Li YR, Zhang Y, Lu Y, Jiang H. Repeated exposure to anesthetic ketamine can negatively impact neurodevelopment in infants: a prospective preliminary clinical study. *J Child Neurol*. 2014;29(10):1333-1338. doi:10.1177/0883073813517508 [PubMed 24659739]
 274. Yeaman F, Meek R, Egerton-Warburton D, Rosengarten P, Graudins A. Sub-dissociative-dose intranasal ketamine for moderate to severe pain in adult emergency department patients. *Emerg Med Australas*. 2014;26(3):237-242. doi:10.1111/1742-6723.12173 [PubMed 24712757]
 275. Yenigun A, Yilmaz S, Dogan R, Goktas SS, Calim M, Ozturan O. Demonstration of analgesic effect of intranasal ketamine and intranasal fentanyl for postoperative pain after pediatric tonsillectomy. *Int J Pediatr Otorhinolaryngol*. 2018;104:182-185. doi:10.1016/j.ijporl.2017.11.018 [PubMed 29287863]

- 276. Youssef-Ahmed MZ, Silver P, Nimkoff L, Sagy M. Continuous infusion of ketamine in mechanically ventilated children with refractory bronchospasm. *Intensive Care Med.* 1996;22(9):972-976. doi:10.1007/BF02044126 [PubMed 8905436]
- 277. Zanette G, Micaglio M, Zanette L, Manani G, Facco E. Comparison between ketamine and fentanyl-droperidol for rectal premedication in children: a randomized placebo controlled trial. *J Anesth.* 2010;24(2):197-203. doi:10.1007/s00540-010-0884-7 [PubMed 20157833]
- 278. Zanos P, Moaddel R, Morris PJ, et al. Ketamine and ketamine metabolite pharmacology: insights into therapeutic mechanisms. *Pharmacol Rev.* 2018;70(3):621-660. doi:10.1124/pr.117.015198 [PubMed 29945898]
- 279. Zeiler FA, Teitelbaum J, West M, Gillman LM. The ketamine effect on ICP in traumatic brain injury. *Neurocrit Care.* 2014;21(1):163-173. doi:10.1007/s12028-013-9950-y [PubMed 24515638]
- 280. Zekry O, Gibson SB, Aggarwal A. Subanesthetic, subcutaneous ketamine infusion therapy in the treatment of chronic nonmalignant pain. *J Pain Palliat Care Pharmacother.* 2016;30(2):91-98. doi:10.3109/15360288.2016.1161690 [PubMed 27092576]

Topic 8588 Version 606.0

Source: UpToDate. Downloaded 2026-05-26.